

Job Market Paper Submission

Isaac Tucker

UCLA Anderson School of Management

This file contains two chapters of my dissertation. The first is an analytical theory paper, *Performance as a Random Variable*. The second, *Playing the Spread: Worker Reliability and Organizational Performance* is its empirical companion. I have split these apart as I prepare them for journal submission, but at conferences and during interviews I have presented them as a single project. I've included both documents here in case the committee finds it useful.

Contact: Isaac.Tucker.phd@anderson.ucla.edu
UCLA Anderson School of Management
110 Westwood Plaza
Los Angeles, CA 90095

Performance as a Random Variable

Isaac Tucker, UCLA Anderson

ABSTRACT

Individual performance is stochastic, but strategic human capital research evaluates workers and policies only in terms of mean or expected performance. This omission embeds strong, rarely stated assumptions about organizations’ ability to absorb bad draws and capitalize on good ones. Using a portfolio model, I show that many small, poor, fragile or interdependent organizations pay a “Volatility Tax” and should prefer reliable workers at any given wage and performance level. Large, well-capitalized, or modular organizations can ignore the spread of performance and focus only on the mean. I develop this insight as Human Capital Portfolio Management and discuss implications for strategy, entrepreneurship, and organizational theory.

MANAGERIAL SUMMARY

Pick two: cheap, good, or never-has-an-off-day. Research has spent decades on the first two (wage against expected performance) but mostly treated the third as noise. I model worker consistency as a separate axis, identify three relevant human resource regimes, and highlight new areas for strategic action. The model inverts typical views on talent: strong firms can chase volatile, high-ceiling talent because they absorb the bad draws, but startups and small businesses must overpay for steadiness or court ruin.

INTRODUCTION

Organizations exist to coordinate the activities of people. And people, alongside other resources and capabilities, determine the strategy, success, and survival of the organization. The challenge, of course, is that human capital has agency (Coff, 1997). Instead of simply holding and trading (as they would a security), organizations must attract, select, hire, incentivize, promote, retain, and develop their workforce. Those that do so more effectively than their competition can achieve both near-term and sustained competitive advantage, which justifies the large and growing body of research on these activities (B. Becker & Gerhart, 1996; B. A. Campbell et al., 2012; Huselid, 1995; Lepak & Snell, 1999).

However, there is at least one important and largely neglected way that human capital behaves just like a financial asset: performance is stochastic. A worker who excels one week may struggle the next. Likewise, a newly promoted high-achiever is likely to see their performance regress towards the mean (Benson et al., 2019; Lazear, 2004). For grocery clerks (Sackett et al., 1988), school superintendents (J. C. March & March, 1977), basketball players (Awtrey et al., 2021), executives (C. Liu & Tsay, 2021; Malmendier & Tate, 2009), and superstar scientists (L. Liu et al., 2018) alike, the difference between one person’s “good” and “bad” performances can be larger than the average difference across individuals (Dalal et al., 2020).

Research has, with few exceptions, attended to only the first moment of this random variable. We study how to raise a worker’s expected performance through oversight, incentives, training, and organizational design, but treat the variance as an error term or behavioral anomaly to be absorbed by the law of large numbers. In practice, workers at a given level of productivity will vary in their performance from period to period. Studies suggest that a worker’s propensity to underperform or overachieve against expectations is a feature of the person-role combination and only loosely correlated with skill-level or expected output (Beus & Whitman, 2012; Dalal et al., 2009; Fleeson &

Jayawickreme, 2015; J. C. March & March, 1977; R. E. Ployhart & Hakel, 1998). In other words, there is an entire dimension of worker performance which is silently assumed away by most human capital strategy work.

Inasmuch as bad days for the person can translate to bad days for the organization, worker volatility presents a material human capital management challenge on par with worker mobility, bargaining, and motivation (Coff, 1997). But how much does it matter for strategy? Certainly, not all organizations are equally exposed. The instinct to simply hire, promote, or retain the “best” worker is likely to be right more often than not. Conversely, we can just as easily imagine a level of volatility or uncertainty in output that would make a manager pine for someone steady, boring, and reliably mediocre.

Human capital strategy needs a set of scope conditions with regard to worker volatility. I approach this problem with a formal model of worker retention in which a decision-maker chooses to retain a reliable worker or replace them with a more volatile, uncertain candidate. I calculate the organization’s specific preference for reliability or exposure to volatility as a function of resources, scale, workforce composition, and production function. The model takes inspiration from asset pricing models and extends readily to other human capital management choices.

Consistent with our understanding of other risky resources or assets, the model suggests that not all organizations are equally exposed. Large, wealthy, and durable organizations can largely ignore individual-scale negative shocks while still capitalizing on positive events. Small, poor, or fragile entities, on the other hand, cannot self-insure against worker volatility and so must pay higher wages, accept lower expected performance, or sacrifice competitive fitness. In other words, rich companies can pay for high average performance. Poor companies must pay for reliability as well. I label this gap between have and have-nots the “Volatility Tax” and show that it neatly divides human capital management into three different regimes with respect to worker volatility. I then extend these findings to more complex and interconnected production environments.

This paper is the first to treat persistent, within-person performance spread as a strategic dimension of human capital. Beyond their value in setting scope conditions for HR/HCM research, these findings have implications for entrepreneurship and the design and management of organizations. I organize these into a research program called Human Capital Portfolio Management (HCPM) and argue that organizations should develop capabilities in three areas: measurement of worker volatility; management of individual volatility with familiar HR interventions; and the design and management of workforce portfolios. Importantly, the implications of Human Capital Portfolio Management are distinct from those found in the study of other risk-return tradeoffs because worker volatility arises from fundamentally different sources and because organizations are uniquely constrained in their interpretation, measurement, and management. HCPM is also an inherently microfounded approach (Felin & Foss, 2005; T. Ployhart, 2011; Puranam, 2018) that can be connected to the literatures on resources/capabilities (Barney, 1991; B. A. Campbell et al., 2012; Coff, 1997; R. E. Ployhart et al., 2014; Sirmon et al., 2007; Teece et al., 1997) and strategy setting (Denrell, 2005; Denrell & Liu, 2012; J. G. March, 1991).

A PRACTICAL EXAMPLE

Consider a stylized managerial dilemma. Worker A is known for their consistency. They come to work every day but never work overtime. They always produce acceptable work, but never anything exceptional. One day, Worker A requests a substantial raise and informs their manager that they will leave the organization unless they receive it.

The manager is confident that Worker A's requested salary would also be sufficient to attract Candidate B. After extensive reference checks, aptitude tests, and structured interviews, the manager is convinced that Candidate B can be expected to produce far more than Worker A in the same role. The choice is obvious, right? For the same wages, the manager would be foolish not to replace the pedestrian Worker A with the exceptional

Candidate B.

But what if Candidate B’s performance is stochastic, drawing from a distribution of possible outcomes? If the role is a creative one, we might frame this as Candidate B mixing great ideas with bad ideas or punctuating a series of very bad designs with moments of brilliance. In a throughput-oriented role like sales, we might think of them as exceeding their quota in some periods while delivering almost nothing in others. Meanwhile, Worker A’s performance is unvarying.¹

With this modification, the choice is no longer obvious. In some periods, Candidate B will underperform Worker A. Depending on the shape of their performance distribution, Candidate B may be a net-negative for the organization for extended periods of time. Candidate B may also produce a volume of work, in a single period, that the organization is unable to absorb. If there are other upstream or downstream functions that depend on this role, they may be unwillingly pulled into Candidate B’s ebbs and flows, even if they themselves are otherwise reliable in their output.

Additionally, Worker A’s performance has been observed, by their manager, over a span of years and so is comparatively better understood. Candidate B performed brilliantly in the interview process, but that represents only a handful of observations. It’s possible that Candidate B is highly reliable and much better than Worker A. It is also possible that they simply had a really “good day” at just the right time to advance their career. Perhaps their focus and motivation were at high-water marks due to their desire for the new role.

Preferring Candidate B now requires the manager to hold three additional beliefs:

1. Candidate B’s interview performance is representative of their mean task performance should they be hired.

¹One important aside: It can be tempting to impose stereotypes and preferences on both workers: Worker A’s performance is boring so they must be an unambitious individual who values consistency and abhors risk; Candidate B is a bit wild, so they must be difficult to manage or more open to variable compensation. This is understandable, but incorrect. Personality differences can impact performance spread, but they are not the same thing. Candidate B could be risk-averse, precisely because they already experience their own boom-and-bust cycles. Or perhaps Worker A’s penchant for risky personal decisions disciplines their work habits. Without clear evidence in either direction, I argue, organizations should manage volatility directly.

2. The organization can weather periods in which Candidate B produces less than Worker A (or nothing at all).
3. Candidate B's output will arrive at a cadence which the organization can convert into value and which does not generate expensive or unsurvivable volatility elsewhere in the organization.

If any of these assumptions do not hold, the organization may (correctly) pass on the exceptional talent and instead pay a premium to retain someone that, by its own estimation, is less talented and productive. The model that follows builds on this example. I focus on retention (and hold mean performance constant) for simplicity, but the implications of the model apply similarly to hiring, benefits, compensation, management, organizational structure, role definition, and other aspects of HCM. We could easily reformat this example for a promotion decision or an internal labor market where information about both candidates is known.

MODEL OVERVIEW

I use a standard definition of worker performance as the aggregated value to the organization of discrete episodes of worker activity (Motowidlo et al., 1997). Each individual performs a set of activities at a point in time in an effort to complete a task. The effectiveness of those behaviors is conditioned by a set of state variables that include resources, firm-specific features, institutions, and the environment. This leaves performance on any given task as a function of the individual's capabilities, their fit with the demands of the task, and their mood, motivation, affect, and attention at the time of the task. And unlike learning curves or age-related maturation trajectories (Bavafa & Jónasson, 2021), volatility is non-directional and typically mean-reverting, suggesting that performance fluctuates above and below a stable personal baseline.

Workers can be low-mean or high-mean and low-volatility or high-volatility and

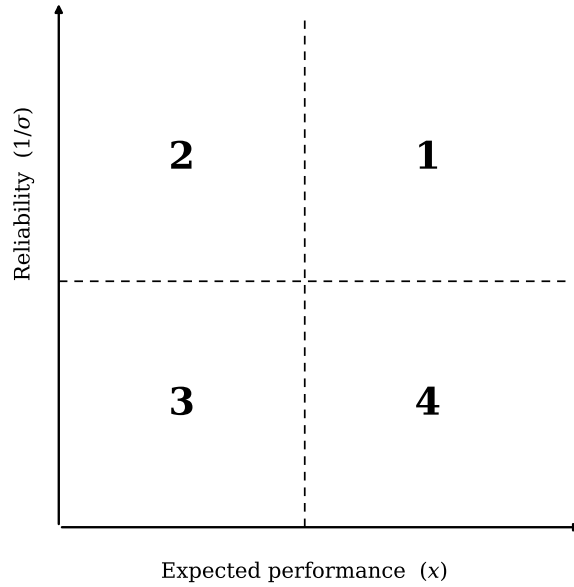


Figure 1: Two dimensions of worker performance

though these are continuous measures, we can develop useful intuition from simple categories. Figure 1 sorts workers into four categories. Quadrant 1 includes workers that are high mean, high reliability. These are superstars and the kind of worker that every organization should hire, retain, and support. Quadrant 3 is the reverse: low-performing, unpredictable, and rarely worth keeping onboard. The remaining two quadrants, which correspond to Worker A and Candidate B from the earlier worked example, present a tradeoff. Is it optimal to hire workers that are great, on average, but have moments of very poor performance? Or would an organization benefit from workers that are not very “good” but provide a stable period-to-period foundation? This tradeoff is also fractal. Within any given quadrant, there should still be gradients of quality and reliability. So even wealthy organizations with the resources to hire “the best” can still benefit from distinguishing between reliable and performant candidates.

In practice, the choice of candidate or quadrant depends on the specifics of the organization, workers, and the labor market. Each team, firm, or manager has a unique set of isoquants which reflect relative value of each type of worker at a given wage. The model

below sidesteps this complexity by setting mean performance and wages equal between two worker types and pricing only the difference in volatility or uncertainty. In effect, this compares Quadrant 1 with Quadrant 4 (high mean workers with differing reliability) and, equivalently, Quadrant 2 against Quadrant 3.

Hereafter, I use the terms performance, productivity, and output interchangeably to denote a worker’s contribution to the organization’s production function. This is a minor abuse of terminology (elsewhere, performance is used to denote behaviors and productivity the resulting value as in J. P. Campbell and Wiernik, 2015) which I justify by noting that the firm’s strategy does not distinguish between the two. The conversion of behaviors into value is an operational challenge, but the choice of employees and the composition of a firm’s human capital portfolio is strategic.

Two Workers

I model an organization’s choice between two workers. The first type is Reliable while the second is Volatile. As in the prior example, I will assume that the Reliable worker is currently employed and that the decision-maker has full information about their type. Each Reliable-type worker produces deterministic and positive productivity for the organization each period: $x_r > 0$. A Volatile-type worker’s performance is stochastic with mean and standard deviation: $x_v \sim (x_r, \sigma)$. Rather than specify a distribution (and to avoid wading into an ongoing debate²), I model volatile performance as a location-scale transform of a fixed, mean-zero shock:

$$x_v = x_r + \sigma z, \quad E[z] = 0, \quad \text{Var}(z) = 1,$$

where z is continuous, has a single-peaked density, and has finite moments. σ is the volatile worker’s standard deviation. Because $E[x_v] = x_r$, the two workers share an expected

²See O’Boyle Jr. and Aguinis (2012), Beck et al. (2014), Spain et al. (2013), and Bradley and Aguinis (2023), among many others.

output and the volatile worker is a mean-preserving spread (Rothschild & Stiglitz, 1970), raising σ moves mass into both tails while holding the mean fixed. Some of the results that follow use one additional regularity condition. Under additive production, the two firms' output distributions (retained and replacement) cross once in each tail. The lower-variance retained state has greater density near the common mean, while the volatile replacement state has greater density in the tails.

The worker's mean and standard deviation (x_r and σ) reflect the manager's beliefs about each worker's future performance in their assigned role. The true parameters of the underlying performance distribution are unknowable. For the retained worker, these beliefs are informed by an extended track record and are comparatively precise. For the replacement candidate, beliefs are formed from a limited set of signals and so carry greater uncertainty. Though I label the replacement candidate as a Volatile type, the model doesn't distinguish between sources of uncertainty. In the mind of the manager, there is no difference between a truly volatile candidate whose performance distribution is correctly estimated and one whose performance is stable but poorly estimated. Both uncertainties fold into σ and the model results can be interpreted in light of either interpretation. An organization that cannot absorb a volatile employee also cannot absorb the uncertainty inherent in hiring a new worker, even if the latter seems to be reliable.

However, for purposes of firm strategy, it is the former interpretation of uncertainty and volatility that I focus on here. There is a substantial body of research addressed to the problem of labor market signaling uncertainty (Altonji & Pierret, 2001; Jovanovic, 1979; Spence, 1973), but it generally assumes that time and repeated observation reduce noise. A "wait-and-see" approach will have no effect on the kind of enduring, work/role-specific volatility addressed here. If anything, the model below suggests a firm that naively assumes that every worker will eventually "settle down" is likely to operate at a disadvantage against its more sophisticated peers.

The Organization

A representative organization has a workforce consisting of these two worker types (N^r, N^v with some x_r and σ) and a production process $Q(N^r, N^v)$ which converts individual worker output into the organization's output. The organization also has some resource position in each period (W_t), net of prior period profits, shocks, and ordinary frictions of business. Resources include the organization's capital, assets, capabilities, reputation, and credit-worthiness (Barney, 1991). Profits carry forward: $W_{t+1} = W_t + Q(N^r, N^v)$. Prices are normalized to one, wages are included in productivity, and all other costs are normalized to zero.

I assume that worker performance variation is independent of other workers and of the firm's resource position. In practice, firm-level conditions can introduce a correlated performance risk which amplifies output variance substantially. If workers' bad days are likely to overlap, the effective volatility of the portfolio is even higher than the model assumes and the reliability premium is larger (so long as the organization is not otherwise insured against left-tail risk). If, as is likely, there is some measure of covariance in worker performance, then the below results represent a conservative lower bound on the importance of performance volatility to strategy.

Baseline features of the model can be interpreted as indications of the quality of the organization or environment. All entities have the same risk tolerance and decision-making quality, but highly productive employees (high x_r) suggest a more effective organization, a more favorable environment, or a better business theory. Likewise, greater wealth (W_t) can be interpreted as an indicator of the quality of the organization or its aptitude with regard to raising capital or attracting talent.

Production Functions

The impact of one person's volatility on the organization depends on how their productivity translates into organization output. If an employee is a single point of failure

for an entire factory, the impact of any error or left-tail draw is magnified. An employee working independently is more limited in the damage that they can do, but the organization still cannot derive outsized benefits from over-performance. Because the way that worker performance combines is, itself, a property of the role and task (T. Ployhart, 2011), I model three production functions, each chosen to illustrate one extreme of independence and inter-connectivity.

Additive production, in which all employees work independently and are perfect substitutes:

$$Q(N^r, N^v) = N^r x_r + \sum_{n=1}^{N^v} x_{v,n}$$

This is a reasonable representation of many small companies as well as some teams. Consider, as examples, a sales organization in which productivity is the sum of the individual agents' sales or a piecework manufacturing facility where output is the sum of the individual workers' productivity. Organizations or teams with additive production are equally sensitive to left and right-tail variance.

Leontief production, in which all workers are interdependent:

$$Q(N^r, N^v) = A(N) \cdot \min(x_r, x_{v,1} \dots x_{v,n})$$

The throughput of the system is equal to the lowest performing individual or station, then scaled by some positive, strictly increasing, unbounded function $A(N)$ such that $A(N+1) > A(N)$ and $A(N) \rightarrow \infty$.

Beyond the obvious applications to assembly-line manufacturing, many small companies also, effectively, operate on this kind of production. Consider a firm with three employees: an engineer, a sales agent, and a consultant/delivery expert. To generate revenue, the engineer must produce a product, the sales agent must sell it, and the consultant must implement it. A bad performance from any one person nullifies the entire organization's output (Kremer, 1993; Perrow, 1984).

The Leontief is the most sensitive to left-tail variance and is unable to extract any value from right-tail variance. It is the opposite of additive production in that workers are perfectly interdependent. In practice, most organizations fall somewhere between these two. For example, organizational decoupling (Orton & Weick, 1990; Weick, 1976), and slack in intra-organization handoffs effectively provide a buffer against events or shocks by allowing two functions to be nominally interdependent but practically disconnected and temporally independent (Bourgeois, 1981; Cyert & March, 1963; Meyer & Rowan, 1977). Likewise, resource redeployability and organizational ambidexterity reduce the costs of production slowdowns elsewhere in the organization (Helfat & Eisenhardt, 2004; O'Reilly & Tushman, 2013; Sakhartov & Folta, 2014).

The effectiveness of these interventions determines where, on the Additive-Leontief continuum an organization falls and how much weight each of the model's respective predictions should receive. Organizations with tightly coupled production or no tolerance for failure should attend primarily to the Leontief results. Those with modular structures will find more value in the additive case. Note that I do not model these intermediate functions directly. This is done for analytical tractability and because these chosen extremes effectively bound a larger space of different production functions. Choosing an interior production function (Cobb-Douglas, for example) yields combinations of these behaviors with ambiguous magnitudes and conflicting signs. As the objective of this project is simply to derive scope conditions, this adds complexity without correspondingly useful insight.

Competitive or Tournament production in which only the best performance within a team or organization matters:

$$Q(N^r, N^v) = A(N) \cdot \max(x_r, x_{v,1} \dots x_{v,n})$$

Within an organization, functions and teams may have different levels of interdependence,

with different production functions at the business unit or organizational level. It is difficult to imagine an entire organization with competitive production, but many individual functions will follow this kind of “sweepstakes” in which the best idea or concept is employed elsewhere in the organization (Boudreau et al., 2011; Dahan & Mendelson, 2001). Consider a lab in which a team of inventors each produce their own prototype: the organization selects the best of the prototypes and puts it into production and the second best idea is scrapped. Competitive production is largely immune to left tail variance (so long as the organization has a single Reliable-type employee, as I will discuss), and benefits fully from right-tail variance.

Competitive production also helps address a category of production in which two or more complementary inputs generate outsized value when paired together (Athey & Stern, 1998). These cooperative or synergistic production functions lie on a continuum between competitive and additive production and share a relative immunity to left-tail variance and ability to capture right-tail variance.

Terminal Failure

Should the organization exhaust its resources, it fails and receives no profits in the future.³ In exchange for taking on marginal failure risk by hiring a volatile worker, the manager requires compensation commensurate with the risk. Because organization size, composition, and features interact to make the organization more durable or fragile, the preference for workers of a certain type varies over time and across settings.

The idea that risk-of-failure warps decision-making and creates a preference for volatility is well-understood in other literatures. Real-options theory shows that volatility increases the value of assets with capped downside (McGrath, 1999). Decision-makers with little to lose take larger risks (Kahneman & Tversky, 1979; J. G. March & Shapira, 1987). Lagging firms invest more heavily (and riskily) in R&D (Cabral, 2003). Fund managers

³This is a standard feature of ecological or chance models in strategy (Denrell, 2004, 2005; Denrell & Liu, 2012; Levinthal, 1991; J. G. March, 1991).

with underwhelming returns take more uncertain bets (Agarwal et al., 2011; Brown et al., 1996). And an extensive body of lab studies suggests that the absence of risk, either because failure is certain or because the agent is playing with house money, leads to a preference for high-upside plays (Thaler & Johnson, 1990). This paper extends that same logic to workforce composition by allowing first period wealth to be negative. The second period, then, becomes the organization’s last-chance to survive.

Hereafter, I refer to this state of exhaustion as “failure” though it need not only represent cases where an organization ceases to function. Instead, this is designed to capture any disproportionately negative outcome for the decision-maker. The failure state embeds two additional assumptions about the organization. First, the consequences of failure (manager termination, firm insolvency, investor displeasure, margin calls, etc.) must apply directly to the decision-making entity. By construction, there is no principal-agent risk (Jensen & Meckling, 1976) in the model. However, where the decision-maker internalizes only a fraction of the organization’s failure penalty, their preference for volatility may be suboptimal for the organization (Coles et al., 2006; Gompers & Lerner, 1999). This is not modeled explicitly but I instead flag it as a caveat to the model results.

Second, left-tail risk and consequences must resolve at a relevant frequency. If worker output is highly volatile at the daily level, but the organization only realizes the consequences when payroll runs every month, then the law of large numbers makes it likely that a bad day will be balanced out before it harms the organization. On the other hand, a sales representative or team with volatile annual performance can cause a promising organization to fail with a sufficiently bad week at year-end. A deal that closes on January 1st is worth substantially less to both the organization and the worker than an otherwise identical sale which is signed at 11:59 pm on December 31st.

The flow of the model is such that the organization hires a single worker, then all workers work, and the organization realizes profits or losses. A company may hire a third worker months or years after their second, but employees 99 and 100 might start on the

same day. For convenience, I do not specify the length of a period, but the model assumes that the performance and profit intervals are identical and non-negligible (Samuelson, 1963). Performance on any time scale can function as an aggregation of many smaller-scale draws, so x_v is the convolution of however many individual draws the worker may make in the same period as the organization realizes its performance. Still, to avoid unsupportable claims, I make no magnitude comparisons across different parameter sets and instead build theory off of comparative statics.

The Manager's Problem

This model differs from Markowitz (1952), March (1991), and other antecedents in its emphasis on a single retention/hiring choice. Portfolio management and exploration-exploitation models optimize the allocation of resources across a set of alternatives and allow for a complete rebalancing of investments in each period. In finance, this is entirely appropriate. An investor whose risk tolerance or liquidity needs change can sell or trade assets in response. However, investments in existing products and/or new product development entail discrete commitments to both workers and fixed assets, and so are more difficult to unwind. A manager cannot costlessly reshuffle their entire headcount spend to new workers.⁴

The state at time t is defined by the organization's wealth and the number of each employee type that it currently employs: (W_t, N^r, N^v) . Resources and profits carry forward according to: $W_{t+1} = W_t + Q(N^r, N^v)$ and the organization fails and liquidates if $W_{t+1} < 0$. The probability of survival is $S(W_t, N^r, N^v) = Pr(W_{t+1} \geq 0 | W_t, N^r, N^v)$.

I collapse all future period payoffs into a positive, bounded continuation value $V(N)$, which represents the value of the organization's workforce conditional on survival. This value captures organization-specific planning horizons and discounting.

⁴To be clear, this is not a criticism of March (1991); his work is concerned first with the firm's posture towards learning and is given proper microfoundations and frictions by Siggelkow and Levinthal (2003) among others. Rather, I note the distinction because human capital is uniquely difficult to re-balance in this way (Coff, 1997).

The manager chooses whether to retain the reliable worker or replace them with a volatile worker with the ultimate objective of maximizing total returns:

$$\begin{aligned} \max \{ & \underbrace{E[Q(N^r, N^v)] + S(W_t, N^r, N^v)V(N)}_{\text{retain reliable worker}} \\ & \underbrace{E[Q(N^r - 1, N^v + 1)] + S(W_t, N^r - 1, N^v + 1)V(N)}_{\text{replace with volatile worker}} \} \end{aligned} \quad (1)$$

The two worker types differ only in the spread of their performance and the resulting impact on survival:

$V(N^r, N^v) = V(N^r - 1, N^v + 1)$ and $S(W_t, N^r, N^v) \neq S(W_t, N^r - 1, N^v + 1)$. To capture this difference, I introduce the Reliability Premium (λ) which is paid out in the initial period. This compensating differential reflects the relative value of a reliable worker compared to a volatile worker in a given state. Formally:

$$\begin{aligned} \lambda = & \underbrace{[S(W_t, N^r, N^v) - S(W_t, N^r - 1, N^v + 1)]}_{\text{difference in survival rates}} \times \underbrace{V(N)}_{\text{continuation value}} \\ & + \underbrace{E[Q(N^r, N^v)] - E[Q(N^r - 1, N^v + 1)]}_{D \text{ (difference in expected output)}} \end{aligned} \quad (2)$$

The first term prices the marginal failure risk the volatile worker introduces. The second, D , is the difference in expected organizational output between the two choices. Because the two worker types share a common mean, D vanishes under additive production. Under interdependent or competitive production the firm's output is nonlinear in worker output, so D takes a definite sign:

$$D \equiv E[Q(N^r, N^v)] - E[Q(N^r - 1, N^v + 1)],$$

$$D \begin{cases} = 0 & \text{additive production (sum)} \\ \geq 0 & \text{Leontief production (min)} \\ \leq 0 & \text{competitive production (max)} \end{cases} \quad (3)$$

Table 1: Model parameters and notation

Symbol	Meaning
N^r	Number of reliable-type workers
N^v	Number of volatile-type workers
N	Total workforce: $N^r + N^v$
x_r	Reliable worker's per-period productivity ($x_r > 0$); also volatile worker's mean (by construction)
x_v	Volatile worker's stochastic performance, $x_v \sim (x_r, \sigma)$
σ	Std. dev. of volatile performance (volatility); also absorbs estimation uncertainty
$Q(N^r, N^v)$	Production function (additive / Leontief / competitive)
$A(N)$	Scale function in Leontief and competitive production
W_t	Resource position (wealth) at time t
W_{t+1}	Next-period wealth; $W_{t+1} = W_t + Q(N^r, N^v)$
$S(W_t, N^r, N^v)$	Survival probability = $Pr(W_{t+1} \geq 0 \mid W_t, N^r, N^v)$
$V(N)$	Continuation value
λ	Reliability premium (compensating differential)
D	Difference in expected output between worker types
t	Period index

Reliability Premium, Volatility Tax

The manager prefers the volatile candidate if the difference in their performance or compensation compared to a reliable employee is, in expectation, greater than the marginal risk of failure multiplied by expected productivity in future periods. This premium could be denominated in wages or indicative of differences in mean performance. It might also represent the value of information about a worker's type.

By construction, there is no reliability premium when the hiring manager’s choice has no impact on survival. If failure is certain, survival is guaranteed, or some other feature of the organization provides insurance against worker volatility

($S(W_t, N^r, N^v) = S(W_t, N^r - 1, N^v + 1)$), λ will be equal to zero or D and the hiring manager will be free to choose the “best” candidate by expected performance.

Organizations would doubtless prefer to have a lower reliability premium, but there will naturally be differences in premiums across entities or firms. This gap, which I label the “Volatility Tax,” represents the disadvantage incurred by organizations that are sensitive to volatility.

Finally, recall that the primary objective of this paper is to determine which organizations are exposed to performance volatility risk. The reliability premium on a marginal worker is sufficient to gauge the direction and magnitude of this risk. A thoughtful reader may immediately jump to portfolio-like organizational implications (hiring, redeployment, ambidexterity, etc.) à la Markowitz or March, but this is outside the scope of this model and reserved for the discussion. Below, I first develop the model using additive production. I then show how findings from this base case evolve in response to more interconnected or competitive production functions.

RESULTS FROM ADDITIVE PRODUCTION

To build intuition, it is helpful to look at the model through the lens of basic probability. As structured, an organization’s wealth in the next period is the current wealth plus the sum of this period’s worker performance draws. With independent workers, that sum is a convolution, and so organizational output is itself a distribution. The manager’s whole problem thus reduces to a simple question of how to reduce probability mass on the wrong side of zero. The manager would prefer to hire the highest-mean worker, regardless of the distribution of performance. However, a more volatile worker fattens the left tail of the organizational distribution. The increased failure risk is priced by the reliability premium.

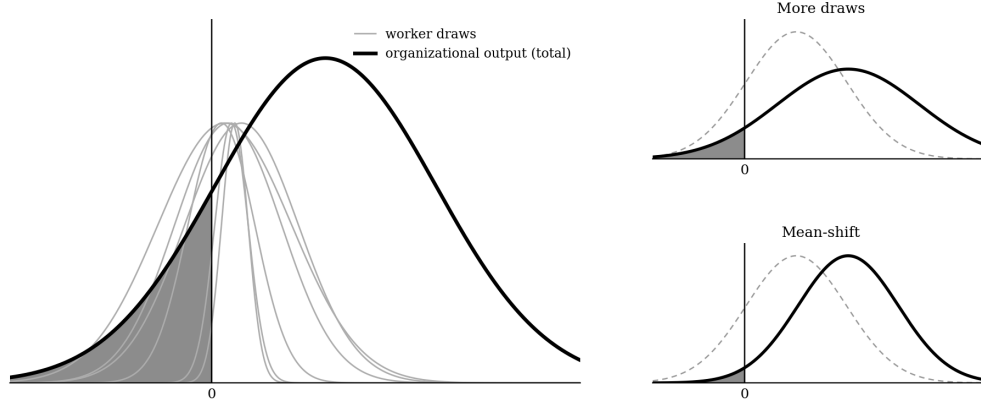


Figure 2: Left Panel - Under additive production, organizational performance is the convolution of worker performance draws.

Right Panel - Organizations can reduce risk associated with stochastic worker performance by adding more draws or by mean-shifting the organization relative to the failure point.

The manager has two levers to reduce the probability of failure, as shown in Figure 2. First, more draws from more workers pull mass from the tails of the organization's distribution to the center via the law of large numbers. Alternately, an increase in wealth or worker productivity shifts the entire distribution to the right, reducing the likelihood that any one worker's bad day can be catastrophic for the organization as a whole.

Wealth and Scale

The reliable worker is risk-free with positive performance. The survival rate $S(\cdot)$ is increasing in W_t and N^r . Given that volatility is uncorrelated across workers, additive production also produces an asymptotically increasing survival rate in N^v , though it grows more slowly than N^r . Continuation value $V(\cdot)$ is monotonically increasing in N .

It follows that increases in the number of employees reduce the reliability premium by increasing the chance of survival. When the organization is sufficiently large, the probability of survival is independent of the choice and so the reliability premium goes to zero, leaving the organization indifferent between worker types on the basis of volatility. Matching the intuition from Figure 2, this is the scale effect:

Proposition 1 (Scale effect). *Under additive production, while*

$S(W_t, N^r, N^v) > S(W_t, N^r - 1, N^v + 1)$, the reliability premium is positive and vanishes as the workforce grows large:

$$\lambda > 0, \quad \lim_{N^r + N^v \rightarrow \infty} \lambda = 0.$$

The same mechanics and intuition apply to resources and wealth. Under additive production, an organization possessing a large stock of valuable resources cannot fail as a result of individual, non-systemic volatility. Let W_A^* denote the wealth level corresponding to the lower-tail density crossing in the output distributions. Beyond W_A^* , the reliability premium is decreasing in resources until the organization accumulates sufficient wealth to ignore volatility altogether. Model features which shift output relative to the failure threshold behave similarly under mild assumptions. This is the wealth effect:

Proposition 2 (Wealth effect). *Under additive production, while $W_t > W_A^*$, the reliability premium is positive and decreasing in resources, and goes to zero as they grow large:*

$$\lambda > 0, \quad \frac{\partial \lambda}{\partial W_t} < 0, \quad \lim_{W_t \rightarrow \infty} \lambda = 0.$$

Volatility as Liability of Smallness

The Volatility Tax has three interpretations. It can represent the gap between sophisticated organizations (those with an understanding of performance as a random variable and the ability to measure it) and naive hiring managers (those who attend only to expected performance). If all parties are aware of a worker's penchant for bad days, the Volatility Tax represents the value of correct information about the spread of a worker's performance. However, most notably, in an environment with full information and sophisticated hiring practices, the Volatility Tax is a structural penalty imposed only on small, poor, and fragile organizations (Propositions 1 and 2).⁵ Calling back to the initial example, a higher volatility tax means that the firm is less able to weather periods in which

⁵The appendix provides a worked numerical example.

a worker produces nothing of value. These are the organizations that should, rationally, choose the unremarkable employee over the superstar, and they are worse for it.

Resource and capability constraints represent a well-documented hazard (Stinchcombe, 1965). Organizations lacking cash reserves, human capital, credit, or reputation are less able to weather shocks, respond quickly to opportunities, or make larger and more efficient investments. The Volatility Tax can be added to these “liabilities of smallness” because it presents resource-constrained organizations with a dilemma in which every alternative carries with it some cost that can only be defrayed by growing large or wealthy. Hire the best employees but risk failure due to one bad day? Retain mediocre workers? Overpay for workers who are neither mediocre nor volatile? Or make costly investments to redesign internal processes and structures? There is no right answer, but organizations that fail to navigate this choice space effectively are likely to disappear.

As a general rule, the liabilities of smallness exert a pull which only organizations that are sufficiently lucky and capable can escape. Those that make it out of the gravity well do so by expanding their resources and capabilities. As a result, the antecedents, mechanisms, and impacts of growth are central to entrepreneurial strategy (DeSantola & Gulati, 2017). The volatility tax represents a novel contribution to this literature, though it also complicates our understanding of growth strategy.

Conventional wisdom suggests that entrepreneurs need to take large risks in order to succeed. Some of this is definitional: every founder wagers that their business can create value that is either overlooked or inaccessible to better-resourced organizations. More materially, scale conveys a compounding advantage. Organizations in a lagging position need to take risks in order to avert a vicious cycle in which the rich get richer and the small and nascent inevitably close up shop (Aghion et al., 2001; Cabral & Mata, 2003; Denrell, 2005; J. G. March, 1991). But startups that play high-variance human capital strategies, are likely to incur the volatility tax, further diminishing the likelihood of success. Worker volatility compounds preexisting challenges in HCM and so is akin to the tyranny of the

Tsiolkovsky rocket equation in which the weight of fuel needed to proceed to space requires exponentially more fuel to lift it to the point where it can be burnt. However, this also suggests that better management of volatility is one path to more effective entrepreneurial strategy.

Wealth as Treatment

Just as gravity wanes with distance, so too do the liabilities of smallness and poorness. When an organization reaches sufficient scale, volatility ceases to be a problem. Compared to smaller organizations, larger organizations can either employ better workers for the same wages or the same workers at a discount. This is consistent with our understanding of assortative matching, which suggests that workers and organizations mutually sort and match with similar quality counterparts (Kremer, 1993). There are a variety of mechanisms, but assortative matching is typically framed as a function of a richer organization being able to pay better wages, make complementary investments, or otherwise become more attractive to candidates by *using* its resources (Abowd et al., 1999; G. S. Becker, 1973; Ouimet & Zarutskie, 2014). Uniquely, in this model, the organization does not pay a premium for talent. Rather, because it has a lower reliability premium, it is more able to absorb and retain talent that is high mean but high volatility. So long as there is sufficient dry powder (in the form of capital, credit, scale, or reputation) to weather small-scale, uncorrelated shocks, the wealthy organization gets better talent without paying for it.

In this, heterogeneous worker performance risk exposure functions like reputation with regard to human capital management. Highly regarded organizations can attract and retain top talent without paying a commensurate premium (Burbano, 2016; Sorkin, 2018) which, in turn, creates a self-reinforcing cycle of sustained competitive advantage.

This also bears directly on our understanding of the role of venture capital in firm success. While there is strong evidence that venture capital helps firms grow (Davila et al.,

2003; Hellmann & Puri, 2002; Kortum & Lerner, 2000; Puri & Zarutskie, 2012), it is sometimes difficult to separate treatment and selection effects (Nanda & Rhodes-Kropf, 2013; Sørensen, 2007; Sorenson & Stuart, 2001). Consistent with evidence that wealth alone can produce better outcomes (McKenzie, 2017), this model suggests that added investment benefits the receiving organization by allowing it to ignore the volatility tax and instead employ better employees than less well-funded competitors.

The implication is that access to capital is even more important for small organizations and that the specific use of funds is important. Whereas venture capitalists or aggressive leaders may strive to be on the frontier of growth, an organization with enough unspent funds to insure against worker volatility (Puri & Zarutskie, 2012) can employ better workers than otherwise. This partially offsets the opportunity cost of running lean and allows these capital reserves to remain as a store of organizational slack and a hedge against other types of risk (so long as these risks are uncorrelated with worker volatility). Unspent funds thus convey some of the benefits of scale without imposing any compensating liabilities.

Many investors contribute not just their capital but also their imprimatur. Taking the broader definition of resources, this suggests that small and new businesses might benefit as much from the implied backstop of a committed investor as they do from the capital itself (Stuart et al., 1999). And because the reserves of a high-profile venture capital fund are functionally unlimited compared to those of a startup, a growth-oriented founder can spend every penny of liquid capital on capable and volatile talent, effectively borrowing against their funders' reputations to hire "better" employees than would otherwise be optimal.

The reverse of this wealth effect suggests that the optimal growth path for a small organization is more conservative than conventional wisdom suggests (Aldrich & Ruef, 2018). No matter how compelling the business idea, an organization with neither cash reserves nor noteworthy investors should be careful to avoid following in the footsteps of its

unicorn forebears and gazelle peers.

The Failing Venture

But if the Volatility Tax incentivizes small organizations to hire reliable workers (Propositions 1 and 2) , why does it *feel* like the opposite is true? We might associate larger organizations with more conservative workers and startups with colorful and eccentric personalities. The staid, Swingline stapler-loving accountant simply doesn't work at a startup and the brilliant ayahuasca enthusiast has no time for the bureaucracy and formality of an established firm.

While no empirical work has examined the volatility of startup workers, there is strong evidence that early hires in small firms are younger, less-skilled, less-educated, and more likely to have been previously unemployed (Coad et al., 2014, 2017; Dahl & Klepper, 2015). How do we square the implications of Proposition 1 with evidence that startups are not hiring purely for expected performance (by most conventional metrics)?

In the model, an organization in a sufficiently precarious position (low wealth, low quality, volatile market, etc.), has negligible marginal failure probability associated with a volatile worker. As organizations receive a payout of zero when they fail (and the entrepreneur/manager can collect on their outside option), there is, effectively, downside protection at high failure rates. A Reliable-type employee has limited value because the organization will likely fail with or without them. A Volatile-type employee's left-tail variance has little downside because the marginal probability of failure is small but the right-tail of the Volatile-type employee offers potential returns and a stay of execution.

In these cases the reliability premium is non-positive. A volatile replacement of equal mean raises the chance of an above-threshold draw and so lowers the premium. With no volatile worker on hand ($N^v = 0$) the premium declines in volatility. With one already in place ($N^v \geq 1$) its marginal response to volatility is negative at $\sigma \rightarrow 0^+$.

Proposition 3 (Imminent Failure). *Under additive production, when*

$S(W_t, N^r, N^v) < S(W_t, N^r - 1, N^v + 1)$, the reliability premium is non-positive, and its marginal response to volatility is negative as $\sigma \rightarrow 0^+$:

$$\lambda \leq 0, \quad \left. \frac{\partial \lambda}{\partial \sigma} \right|_{\sigma \rightarrow 0^+} < 0.$$

For $N^v = 0$ the retained firm is deterministic and $\partial \lambda / \partial \sigma < 0$ for all σ .

So how common is this category of strategies? Such risk-taking behavior is commonly observed in both management and investment decisions and is consistent with the behavioral theory of the firm (Cyert & March, 1963). Organizations with strong prior performance take fewer risks. Those in dire straits take more risks and consequently perform worse, even after conditioning on resources/wealth (Bromiley, 1991; Eisdorfer, 2008; Leland, 1998). Hedge funds take riskier bets as they near the end of a reporting period in which present performance is poor and mutual fund managers whose returns lag behind competitors or peers increase fund volatility in the hopes of resurrecting their performance ranking (Brown et al., 1996). Precarious startups behave similarly (Angus, 2019; Ref et al., 2024).

This model requires that the focal organization be at risk of imminent failure before this kind of risky HCM is optimal, but with minimal loss, we can allow only that the decision-maker believes that they are at risk of failure. The real failure rate for an organization is unknowable, so a boundedly rational manager may conclude that they should hire volatile workers when the opposite is true. The reverse is also possible, though excessively employing reliable workers is unlikely to be a terminal mistake. In either case, optimal worker selection (and, therefore, growth) depends on being able to accurately gauge the quality of a theory of business, which is no small task (Felin & Zenger, 2009).

And for many founders, the belief that failure is always close by may not be entirely wrong. The overwhelming majority of startups will fail (Fairlie et al., 2023), but not before experiencing a brief indemnification from performance volatility risk. In any population,

some number of these organizations will survive thanks to a favorable draw from a less-than-reliable employee. The unexpected stroke of genius can become a part of an organization’s mythology, which, when paired with survivorship bias, would account for the salience of oddballs in any history of high-growth firms. It also makes it difficult to separate a good growth strategy from a very bad (but very lucky) one.

This period of extreme fragility and almost-certain failure is, by definition, fleeting. Organizations either hit on a long-shot or fail and disappear. A startup that escapes disaster actually grows more vulnerable to performance volatility. Only once it has grown sufficiently large or wealthy can it ignore volatility once more, albeit for different reasons (Ref et al., 2024). This is consistent with evidence that capital infusions into startups cause hiring behavior to conform to patterns usually reserved for more established organizations (Dahl & Klepper, 2015; Ouimet & Zarutskie, 2014) and provides another dimension to the treatment effect of venture capital.

Three HCM Regimes

Together, these suggest that human capital management has three distinct regimes with regard to worker volatility, as shown in Figure 3. When the organization’s survival is deterministic (either because failure is impossible or because it is certain), volatility can be ignored altogether. When failure is possible, organizations should price volatility commensurate with the risk it induces. And when failure becomes the more likely outcome should workers deliver on expectations, the organization turns to volatility as a lottery ticket.

The boundary between the liability and lottery-ticket regimes is the state at which replacing a reliable worker with a volatile one of equal mean leaves survival unchanged, $S(W_t, N^r, N^v) = S(W_t, N^r - 1, N^v + 1)$.

Failure impossible	Failure possible	Failure probable	Failure certain
$\lambda = 0$	$\lambda > 0$	$\lambda < 0$	$\lambda = 0$
<i>volatility irrelevant</i>	<i>volatility a liability</i>	<i>volatility an asset</i>	<i>volatility irrelevant</i>

Figure 3: Three HCM Regimes with respect to worker volatility.

THE TAILS OF PERFORMANCE

One of the definitive roles of an organization is to coordinate the efforts of more than one person towards a common goal (Scott & Davis, 2015). And, in fact, the more complex and interconnected an activity, the greater the advantage conveyed by performing it within an organization compared to using market mechanisms. Thus far, workers have been independent and so the organizations' exposure to outlier performance, both good and bad days, was roughly symmetric. Again, some basic intuition from probability serves us well. The minimum is concave so swapping the reliable worker for a volatile one of equal mean strictly reduces expected output. The maximum is convex, so trading a deterministic worker for a volatile one raises its expectation. In other words, each of the two production functions presented here mechanically shaves off one tail of the worker's output, with implications for how the worker's performance is absorbed by the organization.

The Weakest Link and the Left Tail

Organizations differ in their inter-connectivity. In the mildest case, a set of interchangeable specialists benefit from sharing overhead: law firms employ accountants so that they don't have to spend attorney time on invoicing and billing. In a more extreme case, the production of complex products can mean that every role in the plant performs a crucial task without which the semiconductor, lithography machine, or aircraft cannot be

completed. The Leontief production function best represents the latter pole, but also applies to a class of organizations in which the objective function is, itself, the elimination of negative outcomes (air traffic controllers, nuclear power plant operators, etc.) (Weick & Sutcliffe, 2001).

The first and most obvious difference in Leontief or assembly-line production is that volatility of any kind is punishingly expensive. One bad period from one worker can nullify an entire business, and so the organization requires an enormous premium for any risk at all. The larger the invoice, the more important good accounting becomes. The more expensive the machinery, the more important is every step in assembly and quality-assurance.

Under Leontief production, output is $A(N) \cdot \min(x_r, x_{v,1}, \dots, x_{v,N^v})$, so the firm survives only if every worker's output (x_r or x_v) is above some floor: $-W_t/A(N)$. So long as this is true, the volatile replacement can only lower survival chances and the reliability premium is positive.⁶

Proposition 4 (Interdependence). *Under Leontief production while $x_r > -W_t/A(N)$ the reliability premium is positive. As wealth grows, the premium converges to D .*

$$\lambda > 0, \quad \lim_{W_t \rightarrow \infty} \lambda = D \geq 0.$$

This also (mostly) rules out the risk-seeking behavior by failure-prone firms under additive production (Proposition 3). In order for the organization to benefit from right-tail draws, it requires *all* employees to simultaneously over-perform against expectations. Except in very small firms, this is vanishingly unlikely.

Because wealth is independent of the production function, the wealth effect remains: resources make the volatile replacement slightly more palatable. When the firm already

⁶While this threshold is important for completeness, it functionally only matters when organizations are subject to some shock to wealth/resources or the scaling function. Absent those, the fact that the company exists and employs workers is evidence that it is operating with workers whose performance is above this threshold.

employs at least one volatile worker ($N^v \geq 1$), there exists a wealth level W_L^* at which $\Pr(x_v \geq -W_t/A(N)) = N^v/(N^v + 1)$.⁷ The reliability premium is increasing in wealth to that point, then declines towards D . When $N^v = 0$ (the replacement introduces the first volatile worker), the right-hand side is zero, no interior peak exists, and the premium declines monotonically toward D .

Proposition 5 (Interdependence, wealth). *Under Leontief production with $N^v \geq 1$, the reliability premium rises in wealth until $W_t = W_L^*$, then falls to D :*

$$\begin{aligned} \frac{\partial \lambda}{\partial W_t} &> 0 \quad (W_t < W_L^*), \\ \frac{\partial \lambda}{\partial W_t} &< 0 \quad (W_t > W_L^*), \quad \lim_{W_t \rightarrow \infty} \lambda = D \geq 0. \end{aligned}$$

When $N^v = 0$ the interior peak disappears and λ falls monotonically to D .

The wealth result in proposition 5 also holds for improvements to a worker's mean performance (x_r).

However, the scale effect reverses. Larger organizations have more to lose and gain no benefit from “good days.” The larger the workforce, the greater the value of future production. Reliable workers increase the likelihood of survival and volatile workers decrease it, widening the gap and raising the reliability premium. The premium also grows in headcount. Each added reliable worker increases the output that a single bad-day can destroy, so no number of workers ever makes even one volatile worker palatable.

Proposition 6 (Interdependence, scale). *Under Leontief production the reliability premium grows large as the workforce expands:*

$$\lim_{N^r \rightarrow \infty} \lambda = \infty.$$

To any operations scholar, these results are intuitive and obvious. Variability at any

⁷This threshold exists for completeness. Most real-world organizations exist in the region above this point.

station in the production workflow damages throughput and creates periods of idleness/crunch. Early innovations in industrial management were largely directed at mitigating against or eliminating worker volatility to prevent expensive disruptions and bottlenecks (Hopp & Spearman, 2011; Taylor, 1911). Obviously, there are few large organizations that demonstrate this kind of fragility, but this is because those firms already make substantial investments in preventing this kind of cascading failure. These are well-known in operations but human capital management doesn't think of them as "costs" of selecting the wrong worker or expressions of the volatility tax.

The most immediate extension/implication for human capital is that organizations that cannot countenance failure should hire only the most consistent performers. If so, we would expect government agencies, utility companies, and payroll departments to be full of the kind of jobs where showing up and following the manual are the only requirements.⁸

There are further implications for the growth of organizations. Consider a startup founder who must work closely with their second employee. If the first is volatile (a common descriptor of entrepreneurs), then the second should be their opposite. A volatile entrepreneur with a bad business idea and a reliable co-founder can outperform one with an excellent idea and a volatile peer. In fact, if production is highly interconnected, all subsequent workers should be reliable: the premium will only grow as the organization does. At some point early in the organization's lifetime, the founder will need to consider whether it is better to hire reliable workers ad infinitum or instead invest in decoupling each function. The latter is likely to entail a substantial fixed-cost and thus suggests yet another potential treatment effect of venture capital (Hellmann & Puri, 2002).

Insurance and the Right Tail

Finally, I consider cases in which production output is entirely dependent on the single best input. Usually, this kind of competitive or tournament-style production function surfaces in

⁸Empirical corroboration of this hypothesis is outside the scope of this paper.

creative or innovative industries. Only one slogan becomes the brand asset. One actor gets to play the lead. One franchisee gets the territory. And only one invention receives a given patent. Given the non-cooperative nature of this work, it's not uncommon for each idea or entry in the contest to come from a separate organization. Still, many leading firms set up internal R&D functions which follow a similar logic. Some large technology companies, for example, make promotions contingent on shipping the best version of a desired product and in many sales organizations, only the top-performing agent earns the grand prize.

In competitive production, the organization only needs one Reliable-type employee. This worker gives a lower bound to the organization or team's output and indemnifies it from any and all risk of left-tail variance and, thus, failure, liquidation, or exhaustion. As before, there is a resource floor $-A(N)x_r$ above which the organization is guaranteed to survive, so long as $N^r = 1$. Because the single reliable worker can guarantee the organization's survival, their value is a material fraction of the entire business:
 $\lambda = [1 - S(W_t, N^r - 1, N^v + 1)]V(N) + D$. This premium rises as workers grow more volatile, because volatility raises the likelihood of failure without the reliable worker. However, this effect only exists over an interior band of wealth $\underline{W}_C < W_t < W_C^*$. Above W_C^* survival is effectively assured and added volatility only widens the output the reliable worker forgoes. Below $\underline{W}_C$ the failure threshold sits so close to the workers' mean that added volatility barely moves the probability of failure. In either tail the output cost dominates and the premium falls in volatility.

Though the first Reliable-type employee is important, they provide no other value to the organization besides insurance and benchmarking. Consider a team with only volatile workers. A larger pool of volatile workers reduces the probability that the organization fails and increases the expected return (and, thus, the cost of failure). There will be a point at which the probability that the maximum draw is less than the reliable worker's output is negligible, and so insurance adds no value. Likewise, a second reliable employee duplicates the functions of the first, adding no insurance value and occupying a

slot that would otherwise produce positive value for the organization. Therefore, as wealth and scale increase, the reliable worker eventually becomes irrelevant.

Proposition 7 (Insurance). *Under competitive production, while $N^r = 1$ and $W_t \geq -A(N)x_r$, the reliability premium rises with volatility on any interval where the insurance gain exceeds the difference in expected-output, falls with wealth, and is non-positive as wealth or the volatile workforce grow large:*

$$\frac{\partial \lambda}{\partial \sigma} > 0 \quad (\underline{W}_C < W_t < W_C^*), \quad \frac{\partial \lambda}{\partial W_t} < 0, \quad \lim_{W_t \rightarrow \infty} \lambda = D \leq 0, \quad \lim_{N^v \rightarrow \infty} \lambda \leq 0.$$

It follows that a second reliable worker is immediately a liability for the organization and every subsequent reliable worker produces increasingly negative value. This effect is exacerbated as the upside of the volatile alternative increases.

Proposition 8 (One is enough). *Under competitive production, while $N^r \geq 2$ and $W_t \geq -A(N)x_r$, the reliability premium is non-positive, and decreasing in worker volatility and the number of reliable workers.*

$$\lambda = D \leq 0, \quad \frac{\partial \lambda}{\partial \sigma} < 0, \quad \Delta_{N^r} \lambda < 0, \quad \lim_{\sigma \rightarrow \infty} \lambda = -\infty.$$

When organizational output depends only on the maximum draw, the organization places a premium on a worker's potential upside, rather than their expected performance (Proposition 7). In fact, the upside-oriented, tournament-style, R&D-focused organization has almost the exact same worker preference as the doomed startup (Proposition 3). This is not a surprising result. Organizations with highly-skewed payoffs, like venture capital funds, operate on almost the exact same principle when selecting companies (Bali et al., 2011; Cochrane, 2005; W. R. Kerr et al., 2014). Both have an effective cap on organizational downside (albeit for different reasons) and so can gamble on a worker's right-tail paying off.

However, unlike the additive and Leontief production functions, tournament production uses workers of each type in fundamentally different ways. If the volatile worker is a lottery ticket, the reliable worker is an insurance policy. They anchor the organization above the failure point, ensuring that even a perfect-storm of underperformance or bad ideas cannot sink the organization. Even in organizations with an unlimited ability to capture right-tail variance, there is still value in a low-performing and reliable employee. Colloquially, this person is sometimes referred to as the adult in the room (Hellmann & Puri, 2002; Wasserman, 2003).

Beyond insurance, the reliable worker may also make it easier to identify truly exceptional ideas by comparison (Gibbons & Murphy, 1990). Without the benchmark of an unexceptional but reliable worker, how can an organization tell whether a new design is good enough? In finance terms, a reliable worker makes it easier to identify the alpha (Jensen, 1968) of an idea or contribution from a more volatile employee.

Ideally a team would build from scratch with an understanding of volatility. The optimal sequence of workforce expansion for a small, innovative or creative organization will be to hire a single reliable employee as soon as possible, then focus only on the highest-upside available (Proposition 8). The same holds for team composition within larger organizations: almost everyone benefits from at least one level-head.

However, Proposition 7 also suggests that some teams can be built to such a size that insurance results from scale, rather than a balanced portfolio. In this case, no reliability is required or warranted, but the team is vulnerable. While it remains large and coherent, the law of large numbers protects against worker volatility. But should the team be split or otherwise reorganized such that the effective size and span of control shrink, there can be a sudden, material exposure to performance volatility risk. This is slightly counterintuitive: adding another manager should increase oversight and reduce worker volatility but may instead accidentally and invisibly eliminate the primary source of stability for the team's output.

Taken together, these results clearly identify the subset of organizations which should attend to volatility. Large, durable, wealthy organizations can ignore volatility altogether (Propositions 1 and 2). Failing organizations should seek it out (Proposition 3). And companies in between, where survival depends on the choice of worker, will require some compensation before replacing a steady performer with a volatile or uncertain worker. When a larger share of a worker’s contribution is filtered through the performance of peers, the organization’s preference for reliability increases (Proposition 4). And when one person’s good day cures a multitude of ills, the organization will prefer employees with a high ceiling over those with a high floor (Proposition 7).

HUMAN CAPITAL PORTFOLIO MANAGEMENT

In a mean-only world, selection is an optimization problem. Given comparable information and constraints, organizations converge on the same workers because a higher expected contribution is all that matters. Differences in outcomes reflect operational effectiveness (signal extraction, screening cost, speed) instead of strategy (Mortensen & Pissarides, 1999; Spence, 1973; Stiglitz, 1975). Introducing within-person volatility creates mean–variance dilemmas in which the “right” choice depends on organization-specific (wealth, size, internal organization, prior hires, etc.) exposure to downside risk and ability to capture upside. Two rational organizations, facing the same worker pool but different Reliability Premiums, can choose differently and both be right.

This is similar to the classic Becker-Gerhart (1996) model in which human capital management supports sustained competitive advantage by creating a wedge between a worker’s market value and their value to a firm (B. A. Campbell et al., 2012; Huselid, 1995; Lepak & Snell, 1999; Wright et al., 1994). Both rely on the inherent difficulty in imitating complex strategies (Rivkin, 2000), but volatility-informed HR strategy doesn’t require organizations lock employees into a wage below their value or invest heavily in firm-specific human capital to generate persistent performance differences. Rather, value results from

intertemporal smoothing of human capital risk. Even at market wages, an organization that is sophisticated with respect to performance volatility will outperform (or at least outlast) competitors in the long run.

However, where researchers have built a sophisticated and theoretically robust view of firm-specific human capital as a locus for strategic action, there is no comparable canon for stochastic worker performance. I argue that research and practice would benefit from a new research agenda, which I call Human Capital Portfolio Management (HCPM) and define as an approach to human resources and organizational design in which organizations make decisions with an explicit understanding of within-person performance volatility. Workers are not fixed producers of output, but stochastic assets whose realized performance fluctuates over time. Organizations, in turn, hold portfolios of such assets, with the objective of maximizing performance returns while managing exposure to human capital risk. In this, HCPM extends calls to treat human capital not as a single stock but as a portfolio of resources and their combinations (R. E. Ployhart et al., 2014), adding the dimension of within-person performance volatility those accounts set aside.

Though the findings above are largely consistent with other literatures on risk management and stochastic performance (Markowitz (1952) and Roy (1952) in finance; Taylor (1911) in operations), human capital resists management with traditional tools. For one, workers are non-tradeable and the volatility of a workforce is more costly to adjust than a portfolio of financial assets. An investor whose risk-appetite changes (or whose asset mix changes exogenously) can re-balance their portfolio by selling some assets and purchasing others. But a hiring manager cannot easily do the same. For another, workers are mobile and self-interested. A management intervention which suppresses variance but also strips worker agency is likely to lead to slacking, attrition, and the complete loss of whatever human capital investments have been made previously. In other words, it's not enough to be aware of worker volatility. We need researchers to identify the best ways to measure and manage it.

This research agenda has three pillars, each of which corresponds to an important abstraction or assumption made in the model above.

1) Measurement - Organizational/managerial ability to estimate a worker (or candidate's) performance distribution from observations, signals, and prior expertise. Includes both a worker's individual α and β and the within-team variance-covariance matrix.

2) Volatility Management - HR policies and managerial practices which improve individual worker consistency, buffer performance volatility, and enable the organization to capture value from the right-tail of worker performance distributions.

3) Portfolio Management - Design and composition of teams across roles and functions, as well as the machinery to modify the portfolio composition strategically in response to changes in worker characteristics, objectives, and environmental conditions.

Measurement

Perhaps the greatest challenge in implementing HCPM is that volatility is challenging to measure. For organizations and researchers alike, it can be difficult or impossible to distinguish between excellent performance and good luck. Worse, though individual volatility is somewhat stable in relation to peers, workers are not static. They change for better and for worse. Even with some understanding of a worker's baseline quality (though these signals are noisy as well), how should organizations determine whether performance in a period is representative of things to come? Or, without even the benefit of being able to observe prior performance, how should organizations think about hiring? Are there signals of reliability that are not also highly valued signals of quality or expected performance?

If evaluators rely on coarse proxies: tenure in role, uninterrupted employment, punctuality, or milestones like home ownership or parenthood, the drive to find reliable candidates might inject biases into HR practices. For instance, men receive a "fatherhood bonus" because their children are interpreted as a signal of stability and commitment but

women with children face a motherhood penalty, likely due to presumed absenteeism and/or competing priorities (Correll et al., 2007). Likewise, age can be taken as steadiness, ethnicity, accent, or immigration status as unreliability, and frequent job changes as flightiness, regardless of actual performance data. Further research can provide alternative measures/signals and prevent these heuristics from amplifying existing biases.

This seems a daunting empirical challenge, but I should note that it is not radically different from efforts to determine signals of worker quality. The difference is simply that we've had a half-century of sophisticated research on the latter.

One opportunity of note: signals of volatility grow more obvious and will accumulate on a worker's resume over time. Conditional on industry and market dynamics, frequent job/firm/role changes, rapid promotions, geographic mobility, changing careers, and emerging skills may all be indicative of volatility. On the other hand, reliability is likely to look like none of these. Drawing from discussions with hiring managers and my own experience, I would suggest that we begin with signals such as time-in-grade. A highly reliable worker is never bad enough to be fired nor exceptional enough to be promoted. There may also be demographic or socio-economic features which operate on a worker's drive/motivation and so enforce discipline or inject chaos in their work habits. And, of course, researchers with access to firm-specific performance data may be able to find new and creative signals of reliability.

Volatility Management and Human Resources

Traditional performance management assumes that the role of HR interventions is to raise mean performance, reduce underperformance, or align behavior with organizational goals. But if we accept that some portion of observed variability is due to within-person volatility, then a new question emerges: How, if at all, do existing performance management practices affect the volatility of performance?

Management oversight, for instance, is generally assumed to suppress

volatility—though sometimes at the cost of efficiency, autonomy, or innovation. But what of other interventions? We do not know whether performance feedback, coaching, incentives, or formal appraisal schemes reduce volatility, have no effect, or even increase it. The same applies to non-HR levers like team composition, job design, and reporting structure. Can organizations dampen volatility without suppressing performance or do more volatile workers require different performance management systems than their more consistent peers. Are they more responsive to incentives? More sensitive to feedback? Or simply less influenceable overall?

There are also questions about complementarity and role-specific sensitivity. Do volatile and reliable workers interact in productive ways (informal anchoring, peer/social regulation, or division of labor, etc.)? Do reliable coaches and mentors help temper their more volatile charges? If so, performance management becomes as much about team construction as individual oversight. Further, how does volatility shape or constrain performance management across roles and levels? Volatile output from an R&D scientist may be tolerated or even expected. Volatility from a low-level individual contributor might be costless to the organization and easily ignored. But, the same behavior from an auditor, security specialist, or plant manager might be grounds for dismissal. Ultimately, if present systems are not up to the task, it will fall to researchers to identify new processes and best practices to manage both volatility and mean performance.

Portfolio Management and Organizational Design

Finally, a key implication of this paper is that intentionally employing reliable or volatile employees is co-constitutive with strategic organizational design. Organizations choose their internal ordering and boundaries strategically as a way of minimizing costs or maximizing productivity and scale. This model suggests that they should also simultaneously consider the composition of their human capital portfolio in making these choices. The location of a function within the organization – its dependencies,

relationships, management span of control, and coupling to other functions – should determine which kind of employee organizations seek, retain, and promote.

There are also questions about the reverse relationship. Conditional on the workers currently employed and the candidates available, how should organizations design their internal workflows? Are flatter organizations more vulnerable to worker volatility? Should production be modular, rather than tightly coupled? Should volatile workers be shunted to peripheral tasks? Does the existence of volatility bear on the boundaries of the firm? Would it make sense to outsource highly volatile tasks? Does the make-or-buy literature still hold if the resource in question is stochastic labor?

Finally, in considering the needs of the individual, more complications emerge. For one, a reliable employee provides a fundamentally different service to the team from their peers, but they do so unwittingly. Reliability is not a deliberate act, but rather an artifact of an individual’s disposition, skill, and approach to work. An organization with competitive production benefits from this consistency but the value is largely invisible in day-to-day operations and, thus, difficult to manage or compensate. To single out the reliable worker for differential treatment is to threaten the behaviors that the organization values (S. Kerr, 1975). But treating them identically to their more erratic peers can be demoralizing for all involved (Greenberg, 1987). The reliable employee never wins the tournament. By construction, their work is never featured or selected for greater attention and yet they remain a member of the organization in good standing and an example to their peers. Neither employee type is likely to be entirely happy with this arrangement.

Model Extensions

Finally, two important extensions of this model are not explored here but are worthy of a brief discussion. First is the expansion of worker performance volatility to multiple organizations. I have thus far assumed that the manager’s dilemma and, therefore, the Reliability premium is independent of the behavior of other organizations. In practice, the

HCM decisions of large incumbents shape the wages and availability of talent for new entrants. We can imagine a case in which a startup with additive production values reliability, but is unable to afford the wages set by a large incumbent with Leontief production and, therefore, a considerably higher reliability premium. Other organizations also affect the payoff function and the availability of resources. Startups engaging in a winner-take-all race to receive scarce venture capital may have no use for reliability, even if, in a vacuum, the risk-free worker would be the more responsible and rational choice.

A second extension concerns competition within the organization. In practice, teams and business units are neither homogeneous nor independent. Competition for promotions, coalition formation, and internal networks create complex and dynamic incentives which shape the manager's decisions distinct from those recommended by the organization-level reliability premium. As in the multi-organization case, the decisions of other managers directly alter the focal manager's tolerance for volatility within their team and their willingness to take risks with their own career and compensation.

In addition to promising interesting additions and caveats to this model, these extensions are important in developing our intuition around HCPM. It's likely that managers and organizations have some concept, already, of worker volatility, so the development of HCPM as a research agenda depends on the credible theoretical extension of this paper's very simple model.

CONCLUSION

There is something delightfully subversive in the idea that any organization, all else being equal, might rationally and correctly choose to hire a candidate who is simply "worse" than the alternative. It is the human capital management equivalent of backing Aesop's tortoise over the faster hare. And yet, the tortoise does win and the hare's propensity to nap or idle is a feature of its performance distribution. Perhaps in a rematch the speedier racer would be more focused. But what about the third race or the fourth? Would the

hare's focus wane? Would boredom set in? Does the lure of the nap in the shade eventually grow too strong once again? And do oddsmakers take this into account? The tortoise may not win many races, but slow and steady will win some.

Researchers and practitioners alike have a blindspot with regard to performance. Despite evidence to the contrary, we overwhelmingly treat performance as a constant. For managers and entrepreneurs, that blindspot is an opportunity. A portfolio-style strategy that integrates talent selection and retention with organizational design, forecasting, and goal-setting is likely to be more performant and more difficult to imitate. For scholars, this is a call to deepen and broaden our pool of knowledge and better explain how to get the most out of human capital. Put another way: in a world of tortoises and hares, success goes to those who understand that the age-old parable is not a morality tale, but a lesson in building a portfolio for bad days.

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A Definitions and Assumptions

I begin by restating some notation from the model setup. In state (N^r, N^v) , the firm retains the reliable worker. In state $(N^r - 1, N^v + 1)$, the firm replaces that worker with a volatile worker. Total headcount is unchanged at $N = N^r + N^v$. The firm survives the period when $W_{t+1} = W_t + Q(N^r, N^v) \geq 0$, i.e. output clears the threshold $-W_t$, so

$$S(W_t, N^r, N^v) = \Pr(Q(N^r, N^v) \geq -W_t). \quad (\text{A.1})$$

With a volatile worker present ($N^v \geq 1$) output has a density and $\partial S / \partial W_t = f(-W_t; N^r, N^v) \geq 0$. When all workers are reliable ($N^v = 0$), output is $N^r x_r$ and S steps at $W_t = -N^r x_r$. The continuation value satisfies $V(N) > 0$ and $\sup_N V(N) < \infty$, and the scale function satisfies $A(N) > 0$, $A(N + 1) > A(N)$, and $A(N) \rightarrow \infty$.

Definition 1 (Reliability premium). The reliability premium is

$$\lambda = [S(W_t, N^r, N^v) - S(W_t, N^r - 1, N^v + 1)]V(N) + D, \quad (\text{A.2})$$

where

$$D \equiv E[Q(N^r, N^v)] - E[Q(N^r - 1, N^v + 1)]. \quad (\text{A.3})$$

Definition 2 (Volatile performance). Reliable workers produce $x_r > 0$. Volatile workers produce

$$x_v = x_r + \sigma z, \quad E[z] = 0, \quad \text{Var}(z) = 1, \quad (\text{A.4})$$

where z has a fixed continuous distribution with density f_z and distribution function F_z . Volatile-worker draws are independent across workers and independent of W_t .

There is no consensus on the shape of worker performance distributions. Lab and field studies point variously to Gaussian, log-normal, exponential, and heavy-tailed forms. In

practice the exact shape may not matter. It is person-role specific, and few organizations have the data or the analytical capacity to pin it down. I therefore build the model, and state the propositions, to hold across shapes. I specify only that volatile performance is a location-scale family, which yields results that hold whenever more spread widens both tails. This needs one assumption, that two firm-output densities cross at two points (as if both had single peak, not multiple humps), and two lemmas.

Assumption 1 (Additive single crossing). Under additive production, let f_R, f_V be the densities of the retained and replacement outputs $Q_R = Q(N^r, N^v)$ and $Q_V = Q(N^r - 1, N^v + 1)$. They share the mean $\mu_Q = Nx_r$. The densities cross at two points $c_L < c_U$ straddling the mean, with $f_R > f_V$ on (c_L, c_U) and $f_R < f_V$ outside $[c_L, c_U]$.

The difference in expected output, D , depends on the production function. Under additive production $D = 0$, since the two worker types share the mean x_r . Under Leontief and competitive production D is signed, and the sign follows from the curvature of the minimum and the maximum.

Lemma 1 (Curvature). *For any convex g , $E[g(x_v)] \geq g(x_r)$ and $E[g(x_r + \sigma z)]$ is weakly increasing in σ . For any concave g , the inequalities reverse.*

Proof. This follows from Jensen's inequality because $E[x_v] = x_r$. If $\sigma_2 > \sigma_1$, then $x_r + \sigma_2 z$ is a mean-preserving spread of $x_r + \sigma_1 z$, so the expectation of any convex function weakly increases with σ . The concave case is analogous. $\square$

Finally, survival turns on clearing a failure threshold, so the last piece of machinery used in the proofs describes how volatility shifts the chance of clearing a target.

Lemma 2 (Tail monotonicity). *For any target level c ,*

$$\Pr(x_v \geq c) = 1 - F_z\left(\frac{c - x_r}{\sigma}\right). \quad (\text{A.5})$$

Where f_z is positive at $(c - x_r)/\sigma$,

$$\frac{\partial}{\partial \sigma} \Pr(x_v \geq c) = f_z \left(\frac{c - x_r}{\sigma} \right) \frac{c - x_r}{\sigma^2}. \quad (\text{A.6})$$

Thus the tail probability decreases in σ for $c < x_r$, is constant at $c = x_r$, and increases in σ for $c > x_r$.

B Proofs

B.1 Additive production

Under additive production, output is a sum, so the shared worker mean leaves expected output unchanged: $D = 0$ and

$$\lambda = [S(W_t, N^r, N^v) - S(W_t, N^r - 1, N^v + 1)] V(N). \quad (\text{B.1})$$

Proof of Proposition 1 (Scale effect). The scope condition

$S(W_t, N^r, N^v) > S(W_t, N^r - 1, N^v + 1)$ and $V(N) > 0$ give $\lambda > 0$ by (B.1). As

$N^r + N^v \rightarrow \infty$, the $N - 1$ workers common to both states concentrate their combined output around its mean, so replacing one reliable worker with one volatile worker shifts survival by a vanishing amount. With $V(N)$ bounded, $\lambda \rightarrow 0$. $\square$

Proof of Proposition 2 (Wealth effect). For $N^v \geq 1$, differentiating (B.1) with

$\partial S / \partial W_t = f(-W_t; \cdot)$ gives $\partial \lambda / \partial W_t = [f_R(-W_t) - f_V(-W_t)] V(N)$. Above $W_A^* = -c_L$,

Assumption 1 gives $f_R(-W_t) < f_V(-W_t)$, so $\partial \lambda / \partial W_t < 0$. With $\lambda \rightarrow 0$ as $W_t \rightarrow \infty$, $\lambda > 0$ there.

For $N^v = 0$ the retained firm's output is the fixed $N^r x_r$, so it survives for certain above

$W_A^* = -N^r x_r$. There $\lambda = [1 - S(W_t, N^r - 1, 1)] V(N)$ is positive, falls in W_t

($\partial \lambda / \partial W_t = -f_V(-W_t) V(N) < 0$), and tends to zero. $\square$

Proof of Proposition 3 (Imminent failure). If $S(W_t, N^r, N^v) < S(W_t, N^r - 1, N^v + 1)$, then (B.1) and $V(N) > 0$ imply $\lambda \leq 0$. In this region the threshold $-W_t$ exceeds the common mean $\mu_Q = Nx_r$.

For $N^v \geq 1$, differentiating (B.1) in σ gives

$$\frac{\partial \lambda}{\partial \sigma} = [f_R(-W_t) - f_V(-W_t)] \frac{-W_t - \mu_Q}{\sigma} V(N),$$

so $\partial \lambda / \partial \sigma$ has the sign of $f_R(-W_t) - f_V(-W_t)$. By Assumption 1, $f_R < f_V$ above the upper crossing c_U . The outputs are location-scale in σ , so $c_U = \mu_Q + \sigma a_0$ for a fixed $a_0 > 0$ and $c_U \rightarrow \mu_Q$ as $\sigma \rightarrow 0$. Since $-W_t > \mu_Q$, small enough σ makes $-W_t > c_U$, so $f_R(-W_t) < f_V(-W_t)$ and $\partial \lambda / \partial \sigma < 0$.

For $N^v = 0$, the retained output $N^r x_r = \mu_Q$ is deterministic, so $S(W_t, N^r, 0) = 0$ throughout the region and constant in σ . The replacement survives if and only if $x_v \geq -W_t - (N^r - 1)x_r$, a threshold above x_r , so $S(W_t, N^r - 1, 1)$ rises in σ by Lemma 2. Hence $\partial \lambda / \partial \sigma < 0$ for all σ . □

B.2 Leontief production

Under Leontief production,

$$Q(N^r, N^v) = A(N) \min(x_r, x_{v,1}, \dots, x_{v,N^v}).$$

The minimum is concave in the replaced draw, so $D \geq 0$ (Lemma 1), and $D > 0$ for $\sigma > 0$ since the spread strictly lowers the expected minimum. Let

$$\tau \equiv -W_t / A(N), \quad p \equiv \Pr(x_v \geq \tau).$$

If $x_r \geq \tau$, the retained firm survives if and only if all N^v volatile workers clear τ , and the replacement firm survives if and only if all $N^v + 1$ volatile workers clear τ . Therefore

$$\lambda = V(N)p^{N^v}(1 - p) + D \quad (x_r \geq \tau). \quad (\text{B.2})$$

Proof of Proposition 4 (Interdependence). By (B.2), λ is a nonnegative survival term plus $D \geq 0$ (above), so $\lambda \geq 0$. For $\sigma > 0$, $D > 0$, so $\lambda > 0$. As $W_t \rightarrow \infty$, $\tau \rightarrow -\infty$ and $p \rightarrow 1$. The survival term $p^{N^v}(1 - p)$ goes to zero, leaving $\lambda \rightarrow D \geq 0$. $\square$

Proof of Proposition 5 (Interdependence, wealth). Since D does not depend on W_t , differentiating (B.2) gives

$$\frac{\partial \lambda}{\partial W_t} = V(N)p^{N^v-1}[N^v - (N^v + 1)p]\frac{\partial p}{\partial W_t}. \quad (\text{B.3})$$

Higher wealth lowers τ , so $\partial p / \partial W_t \geq 0$. For $N^v \geq 1$, the sign of (B.3) is positive when $p < N^v / (N^v + 1)$ and negative when $p > N^v / (N^v + 1)$. The premium therefore peaks at W_L^* satisfying

$$\Pr(x_v \geq -W_L^*/A(N)) = \frac{N^v}{N^v + 1}.$$

As $W_t \rightarrow \infty$, $p \rightarrow 1$, the survival term vanishes, and $\lambda \rightarrow D \geq 0$. If $N^v = 0$, the survival term $1 - p$ decreases in wealth, so no interior peak exists and λ falls monotonically to D . $\square$

Proof of Proposition 6 (Interdependence, scale). By (B.2), $\lambda = V(N)p^{N^v}(1 - p) + D$. The survival term is bounded because $V(N)$ is bounded. The expected-output term is $D = A(N)\delta$, where δ is the difference between the expected minima in the retained and replacement states before scaling by $A(N)$. For fixed N^v and $N^r \geq 2$, δ does not depend on N^r and, for $\sigma > 0$, is strictly positive (the strict $D > 0$ above). Since $A(N) \rightarrow \infty$, $D \rightarrow \infty$. Therefore $\lambda \rightarrow \infty$ as $N^r \rightarrow \infty$. $\square$

B.3 Competitive production

Under competitive production,

$$Q(N^r, N^v) = A(N) \max(x_r, x_{v,1}, \dots, x_{v,N^v}).$$

The maximum is convex in the replaced draw, so $D \leq 0$ (Lemma 1), and $D < 0$ for $\sigma > 0$ since the spread strictly raises the expected maximum. Retain the previously introduced notation $\tau = -W_t/A(N)$ and $p = \Pr(x_v \geq \tau)$. If $x_r \geq \tau$ and $N^r = 1$, the retained firm survives for certain, while the replacement firm fails only if all $N^v + 1$ volatile workers fall below τ . Thus

$$\lambda = V(N)(1 - p)^{N^v+1} + D \quad (N^r = 1, x_r \geq \tau). \quad (\text{B.4})$$

If $x_r \geq \tau$ and $N^r \geq 2$, both states retain at least one reliable worker, so both survive for certain and

$$\lambda = D \leq 0. \quad (\text{B.5})$$

Proof of Proposition 7 (Insurance). For $N^r = 1$ and $x_r \geq \tau$, equation (B.4) gives

$$\frac{\partial \lambda}{\partial \sigma} = -(N^v + 1)V(N)(1 - p)^{N^v} \frac{\partial p}{\partial \sigma} + \frac{\partial D}{\partial \sigma}.$$

Since $\tau \leq x_r$, Lemma 2 gives $\partial p / \partial \sigma \leq 0$. Lemma 1 gives $\partial D / \partial \sigma < 0$. Hence the premium rises with volatility exactly when

$$(N^v + 1)V(N)(1 - p)^{N^v} \left(-\frac{\partial p}{\partial \sigma} \right) > -\frac{\partial D}{\partial \sigma}. \quad (\text{B.6})$$

When nonempty, the set of wealth levels satisfying (B.6) is denoted $(\underline{W}_C, W_C^*)$. At the floor $\tau = x_r$, Lemma 2 gives $\partial p / \partial \sigma = 0$. As $W_t \rightarrow \infty$, $p \rightarrow 1$ and $(1 - p)^{N^v} \rightarrow 0$. Thus the insurance gain vanishes at both ends of the region. If (B.6) never holds, $\partial \lambda / \partial \sigma < 0$ throughout.

For wealth, D is constant in W_t , and

$$\frac{\partial \lambda}{\partial W_t} = -(N^v + 1)V(N)(1 - p)^{N^v} \frac{\partial p}{\partial W_t} \leq 0,$$

with strict inequality wherever the volatile output has positive density at τ . As $W_t \rightarrow \infty$, $p \rightarrow 1$ and $\lambda \rightarrow D \leq 0$. For fixed $p < 1$, as $N^v \rightarrow \infty$, $(1 - p)^{N^v+1} \rightarrow 0$, so again $\lambda \rightarrow D \leq 0$. □

Proof of Proposition 8 (One is enough). If $N^r \geq 2$ and $x_r \geq \tau$, both states contain at least one reliable worker, so both survive for certain. The survival difference is zero, and (B.5) gives $\lambda = D \leq 0$.

Only D moves with σ . Since the maximum is convex in the replaced draw, Lemma 1 gives $\partial D / \partial \sigma \leq 0$, strictly for $\sigma > 0$, so $\partial \lambda / \partial \sigma < 0$. For headcount, write $D = A(N)\eta$, where

$$\eta = E \max(x_r, x_{v,1}, \dots, x_{v,N^v}) - E \max(x_r, x_{v,1}, \dots, x_{v,N^v+1}).$$

For $\sigma > 0$, $\eta < 0$ (the spread strictly raises the expected maximum). For fixed N^v , η is independent of N^r , while $A(N)$ is strictly increasing. Hence $\Delta_{N^r} \lambda = \Delta_{N^r} D < 0$. Finally, as $\sigma \rightarrow \infty$, the expected volatile maximum diverges, so $\eta \rightarrow -\infty$ and $\lambda \rightarrow -\infty$. □

C Numerical Illustration

The results are not Gaussian-specific. This illustration fixes a Gaussian volatile law only to put numbers to the reliability premium. Workers are paid \$100,000 per period. Their expected output is worth \$105,000, so each clears $x_r = \$5,000$ net. A volatile worker's performance is normal around that mean with standard deviation \$30,000. The firm holds no reserves ($W_t = 0$), faces no resource shocks, and has continuation value equal to current expected output.

Consider a firm with two reliable workers. Its output is certain at \$10,000, so it survives.

Replacing one reliable worker with one volatile worker leaves expected first-period output unchanged at \$10,000, but output is now normal with standard deviation \$30,000. The failure probability is

$$\Phi\left(\frac{0 - 10,000}{30,000}\right) = \Phi(-1/3) \approx 0.37.$$

The loss runs through survival, not expected output. The firm keeps its continuation value with probability about 0.63, so expected continuation value falls to about

$\$10,000 \times 0.63 = \$6,300$. Keeping the worker reliable is therefore worth about \$3,700.

The premium falls with reserves and scale. With \$10,000 on hand, the failure probability falls to $\Phi(-2/3) \approx 0.25$, and the premium is about \$2,500. In a firm of twenty reliable workers with \$20,000 on hand, replacing one reliable worker with one volatile worker leaves failure all but impossible, $\Phi(-4) \approx 0.00003$, and the premium is about \$3.

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In writing the manuscript, I used generative AI tools for proofreading, editing, and image generation only. In writing the appendix, I used Claude, ChatGPT, and Refine.ink to construct and test proofs.

Playing the Spread: Worker Reliability and Organizational Performance

Isaac Tucker

Abstract

Workers have good days and bad days, but human capital management research overwhelmingly focuses on mean performance. This paper examines whether organizations recognize and price worker volatility when they can measure it, and whether they should. Using data from professional basketball, I document both wage and production penalties for volatile workers. After separating persistent volatility from playing-time noise, more volatile workers earn less on the margin where contracts are negotiable. A difference-in-differences design then exploits the quasi-random timing of absences as an identifying shock. Holding mean performance fixed, losing a reliable worker inflicts a larger penalty on organizational outcomes than losing a volatile one. I show that the steadiest workers are eighty percent as costly to lose as a top-quintile player (near-stars through superstars), at forty percent of the salary. Though the production benefits of reliability are contemporaneous, the wage premium exists only for workers with a track record of consistency. Because organizations could plausibly understand a worker's consistency before the market does, this gap represents an opportunity for competitive advantage. I close by arguing that the management of worker volatility is a portfolio problem.

1. Introduction

On June 11, 2013, the Miami Heat lost in Game 3 of the National Basketball Association (NBA) finals by 36 points. LeBron James – the league’s most valuable player, in the midst of perhaps the greatest individual season in basketball history, leading a super-team with the NBA’s best record – scored 15 points on 21 shot attempts. It was his worst performance of the season. Advanced statistics suggest he played at a level undeserving of a roster slot (Myers 2020). With the benefit of hindsight, we know that this was an aberration: the Heat would go on to win the best-of-seven championship series in Game 7 behind 37 points from their superstar. But, in all likelihood, there was at least one person among Game 3’s 14 million television viewers who had never watched professional basketball before. A neophyte, lacking context, might reasonably have concluded that one of the greatest players of all time was a mediocre talent on an over-matched team.

Though absurd in the context of professional basketball, the measurement and management of stochastic performance is a material and under-addressed challenge. Organizations routinely evaluate workers on singular or sparse events, despite evidence that these are noisy signals with limited predictive value. Hiring, retention, termination, promotion, and performance bonus decisions respond to periods of recent over-performance and punish underachievers (Lampe & Pance 2020). And yet, bad days are all too common. Whether because the baby won’t sleep, the weather won’t cooperate, or the muse won’t sing, individual performance is subject to nadirs at varying scales and intervals. Meta-analyses suggest that roughly half of variation in performance outcomes in both lab and field settings is attributable to within-person volatility, rather than between-person differences (Dalal et al. 2020; Podsakoff et al. 2019).¹

However, research on organizations almost exclusively focuses on the individual worker’s mean or the dispersion of worker means across a workforce or organization. This omission is particularly important for strategic human capital management. Unlike other assets, human capital

¹In the case of professional basketball, 76–92% of total game-over-game variation is within player-season. Specifically, this is variation in performance among NBA starters (639 players, 143,064 player-games, 2011–2023) for rate-adjusted Game Score (76%), BPM (86%), or plus/minus (92%).

is embodied in workers who firms do not own, who can leave of their own volition, and who can bargain over the value they create (Coff 1997, 1999). Decision-makers must identify which worker features create value for the organization and determine how those characteristics are valued on the labor market. Where there is a gap in the private and public valuations of a worker, the organization can extract value and build a sustainable competitive advantage. Traditionally, this is the result of bundles of HR practices or the development of firm-specific human capital (Becker & Gerhart 1996; Bidwell 2011; Chadwick & Dabu 2009). If the spread of a worker's performance has value distinct from the mean, it would represent an additional (and novel) wedge between labor-market price and organizational value.

This paper examines whether there is a labor market premium for reliable workers, whether worker volatility matters for organizational performance, and whether there exists a gap between the two from which a strategic and well-informed organization might extract an advantage. I answer these questions using data from the National Basketball Association, one of the few research settings in which competition occurs under a set of standardized rules and objectives and performance data is standardized, plentiful, and public.

I first show that organizations price consistency among workers whose compensation is negotiated rather than fixed by the league's minimum and maximum salary rules. There is also some evidence for sequential evaluation: organizations first pay for mean performance, then bargain over reliability at the margins. I then exploit the quasi-random timing of worker absences to estimate the corresponding production value of reliability. Holding mean performance fixed, losing a more reliable worker imposes a larger penalty on team outcomes (point differential and win probability) than losing a more volatile one. The implied wage and production tradeoffs are consistent, suggesting that organizations price reliability because it contributes to performance.

Though the production benefits of reliability are contemporaneous, the wage premium exists only for workers with a track record of consistency. Because organizations could plausibly understand a worker's consistency before the market does, I argue that this gap represents an opportunity for competitive advantage. Further, the top quintile of reliable workers produce

organizational value commensurate with the value generated by superstars at a fraction of the wages. In a setting with substantial resource constraints, this surplus can be reinvested elsewhere for even greater returns, so organizations that excel at identifying reliable workers should be able to consistently outperform those that do not.

1.1. Whither Stochastic Performance?

Since 2000, six top management journals² have published 316 empirical papers containing any mention of human resources or human capital management alongside any mention of performance or productivity³. This corpus contains a wealth of measures and methods, but not a single paper using individual, within-person, intertemporal variance as a focal outcome or treatment variable. This is not to say that the role of volatility is unknown. An adjacent tradition in industrial and organizational psychology treats performance variability as a distinct dimension of worker performance (Sackett et al. 1988) and finds that it interacts with compensation (Awtrey et al. 2021; Barnes & Morgeson 2007). Some authors acknowledge the absence of volatility in their analysis (De Figueiredo et al. 2013). Others include it in their theoretical specification but not their empirical findings (Aral et al. 2012; Battiston et al. 2025; DeVaro 2006).⁴ It is possible that this is a costless omission. However, stochastic performance can impose a non-negligible (Lazear 2004) and sometimes catastrophic (Perrow 1984) cost on the organization. The extent to which these costs impact organizational performance and, therefore, strategy is largely unexplored.

Recent formal work sets out scope conditions for worker volatility and strategy. Tucker (2026) represents worker performance as a distribution, rather than a scalar, and shows that the organization's preference for volatility depends on scale, resources, workforce composition, and internal organization. Two organizations can value the same mean performance very differently, depending on the volatility of the worker. Two workers with the same mean can generate very

²AMJ, AMR, ASQ, Management Science, Organization Science, SMJ

³On Web of Science, the query is (human resources OR human capital) AND (productivity OR performance)

⁴Only a handful of researchers concerned with the effectiveness of healthcare operations have assessed the effect of policies, interventions, and experience on worker volatility (Bavafa & Jónasson 2021, 2024; Lu et al. 2022). However, because this work is addressed to the operations management literature, it does not show up in the initial literature search and may not be readily apparent to HR researchers or practitioners.

different value, depending on how the tails of their performance distribution interact with the organization. At the limit, the best worker in the world might be the wrong person for the job if the organization can't survive long enough to take advantage of their sporadic brilliance. This approach implies two distinct quantities: an organization-specific valuation for a worker's performance (both mean and distribution) and a labor market valuation for the worker's observable performance (plus the effects of bargaining, signaling, and other pressures).

I estimate and compare both. In addition to providing empirical support for the idea that worker volatility is strategic, this also offers insight into the somewhat startling absence of worker volatility from research on strategy and organizations. The gap is consistent with at least three candidate explanations. Perhaps worker volatility simply doesn't matter to organizational outcomes. It might be priced in to wages already. Structures and routines might provide buffering and insulation (Orton & Weick 1990; Weick 1976). Or the law of large numbers might provide most organizations with statistical insurance against extreme draws. If this is the case, one would expect to find evidence that worker volatility has a null effect on organizational performance or is almost perfectly priced across organizations.

We might instead take the lack of attention paid to worker volatility as evidence of a managerial (and academic) blind spot. After all, individuals are known to struggle to separate signal from noise or draw the correct conclusion from stochastic signals (Aldrich & Ruef 2018; Benson et al. 2019; Denrell 2005; Denrell et al. 2023; Lagziel & Lehrer 2019; Lazear 2004; Liu & Tsay 2021; March & Sutton 1997).⁵ This would suggest no wage premium and a material organizational benefit to employing reliable workers.

Finally, an organization-specific reading of Morgan's Canon⁶ suggests that we not attribute to either strategy or bias what can be better explained by measurement challenges. That is: the

⁵The best-documented case is promotions and the "Peter Principle," which observes that newly promoted employees are likely to underachieve at some point because they were promoted on the basis of a period of unusually high output (Benson et al. 2019; Lazear 2004). The Peter Principle is widely known among practitioners and has strong and consistent support from researchers. However, there has been no effort to extend this understanding to the broader area of Human Capital Strategy.

⁶"In no case is an animal activity to be interpreted in terms of higher psychological processes if it can be fairly interpreted in terms of processes which stand lower in the scale of psychological evolution and development."

shape and tails of a worker’s performance distribution are considerably more difficult, expensive, and time-consuming to measure than the mean. It follows that there is value in testing the role of worker volatility in a setting where performance measurement is unusually manageable.

2. Research Setting

Professional basketball has a long history as an empirical setting for questions about performance and randomness. Since Gilovich et al.’s (1985) foundational paper on shooting sequences, a large literature in psychology and sports analytics has used NBA data to study whether player output is more or less random than intuition suggests. Unlike that research, which uses the possession (really, the shot) as the unit of analysis, I examine performance at the level of a single game. One game gives a player ample opportunity to impact the outcome whereas a single possession can occur without the involvement of 9 out of 10 players in the game. Game performance has an advantage over season-level performance in that every game has a winner, a loser, and a common set of environmental and temporal conditions under which it is played. Still, this is a choice made (in part) for convenience. Volatility occurs from moment to moment (Dalal et al. 2020) and every period is the convolution of these micro-events. As our understanding of performance grows, it should naturally direct more attention to the scale at which workers are measured.

2.1. The Benefits of Professional Basketball as a Research Setting

Professional basketball offers three advantages for the study of worker performance and organizational outcomes. First, all organizations and workers are subject to a set of common rules and share (ostensibly) the same definition of success (Aversa et al. 2025). While there is certainly between-organization variation in resources, planning horizons, risk tolerance, and skill/sophistication, incentives broadly push all teams in the direction of winning games and championships within a reasonable time frame. Even cases of “tanking” or losing games in order to obtain roster-building advantages in subsequent seasons are motivated by the eventual desire to win.

The second and third benefits derive from the fact that professional sports have rich and

standardized performance data (Downs et al. 2025). Whereas many organizations are forced to hire, promote, or fire workers based on only a handful of observed data points, professional basketball teams can make decisions in an environment where every pass, shot, and step is logged. This greatly reduces the concern that decision-makers simply cannot see the distribution of a worker's performance. Were we to observe promotion decisions made in a less information-rich environment, we would first have to question whether or not the decision-maker was even aware of the full breadth of their employee's performance. And, as discussed, we would have every reason to think that they were not.

Finally, this wealth of data also allows researchers to partially observe the effort a player is exerting. In an oppositional game, more valuable actions are more costly. In the modern NBA, this means that the highest-value locations on the floor require the most energy to reach. A player seeking to shoot from 40 feet away can do so at any time (with the blessing of the opposing team). Higher-value shots (at the rim or from the free-throw line) require that the player be willing to work harder and, in all likelihood, absorb physical punishment in exchange for their shot attempt. Over the course of a season, a player's willingness to run themselves to exhaustion or be elbowed/shouldered somewhere soft will naturally ebb and flow, which is reflected in the type of shots they seek out.

2.2. Measuring Performance in Professional Basketball

Professional basketball teams play 82-game seasons. Each game lasts 48 minutes and the court is shared by five players on each team, giving 240 team-minutes per game. Teams take turns attempting to score baskets. An average game sees just over 100 possessions per team. On each possession, players can contribute to the success or failure of their team's attempt to score in multiple ways. Many of these are captured and tracked with basic counting statistics. In addition to points scored, players are measured on their shooting percentage (shots made divided by shots attempted), assists (passes which lead to teammate scores), blocks and steals (defensive actions which prevent shooting opportunities for the other team), turnovers (possession lost without a shot attempt) and rebounds (ball recoveries after a missed shot).

As in many other jobs, basketball players have different strengths and inclinations, which frustrates attempts to collapse these statistics into a single number. Point guards, responsible for distributing the ball to others and chosen for their passing ability, will naturally get more assists than players who are on the court because they are tall enough to grab rebounds. The simplest approach, and the one I use in this paper, is a weighted sum of the player's activities called Game Score (Hollinger 2003). The idea behind Game Score is that each possession generates, on average, about one point. So an action which earns the team another possession is worth about a point while one that loses possession subtracts a point. These fractional "points" are then added to the player's point total, which gives their Game Score. The full formula is:

$$\begin{aligned} \text{Game Score} = & \text{Points Scored} + (0.4 \times \text{Field Goals Made}) - (0.7 \times \text{Field Goal Attempts}) \\ & - (0.4 \times (\text{Free Throw Attempts} - \text{Free Throws})) + (0.7 \times \text{Offensive Rebounds}) \\ & + (0.3 \times \text{Defensive Rebounds}) + \text{Steals} + (0.7 \times \text{Assists}) + (0.7 \times \text{Blocks}) \\ & - (0.4 \times \text{Personal Fouls}) - \text{Turnovers}. \end{aligned}$$

Game Score is not a perfect summary performance measure as it omits context (opponent quality, pace of play, location of game, quality of teammates, etc.). However, for the study of game-level performance variance it has several important features. Game Score is universally available, easily interpretable from game to game and has obvious face value. It penalizes bad things, rewards good things, and captures the many different ways to add value as a professional basketball player. A player whose Game Score falls from 30 to 5 in back-to-back games had one good game and one bad game, regardless of how they accumulated those statistics.

In order to ensure that the pieces of performance that are not measured in Game Score do not materially alter the below results, I also make use of one additional "advanced" statistic in the form of Box Plus Minus (BPM). This regression-based Game Score analog takes its name from plus/minus, the team's point differential while a player is on the floor, and estimates how much a player contributes to the team's margin of victory, on average, across an entire season (Myers 2020).

It includes much of the context that Game Score lacks and so is valuable for long-run and head-to-head comparisons. For individual games and the estimation of game-over-game performance volatility, BPM can be trickier to explain or interpret as it includes a more complex set of weights than those in Game Score. Where it is necessary to control for long-run player quality, I use BPM. Otherwise, I use BPM as a robustness check for Game Score based results.

The other problem with Game Score is that it is a counting statistic, so game-over-game variation is a function of both player characteristics and minutes played. This is akin to binomial noise. Consider, for example, that a single shot attempt has an average success rate of 46%. A player who takes 15 shots per game has a per-game sampling error of roughly 13 percentage points. At the league average of 9 shot attempts per game, this grows to roughly 16 percentage points (nearly three times the between-player standard deviation in shot percentage after correcting for this binomial noise). Prior work using NBA data (Awtrey et al. 2021; Barnes & Morgeson 2007) makes no correction or adjustment for this noise and so misspecifies the volatility of player performance.

Subsequent analyses use a minutes-adjusted Game Score standard deviation as an estimate of player-specific variance, net of the mechanical variance associated with limited playing time. For the median player-season (23 minutes per game), this correction removes roughly half of the observed variance. I first normalize Game Score per 36 minutes, which is conventional for NBA statistics and reflects the typical regular-season workload for a top player. This is necessary for comparison across players with different workloads, but it also magnifies the impact of variance. A particularly lucky 3 minute stint can look historically good (or bad) when multiplied by 12. I then decompose the variance of per-36 Game Score across games within a season into a player-specific intercept and accompanying noise:

$$\text{Var}_{\text{obs},i}(GS) = \underbrace{\sigma_{\text{true},i}^2}_{\text{player volatility}} + \underbrace{k \cdot \mathbb{E}[1/m_{i,g}]}_{\text{noise}}. \quad (1)$$

Here, $m_{i,g}$ is minutes played by player i in game g , and $k \cdot \mathbb{E}[1/m_{i,g}]$ is finite-exposure noise that shrinks as playing time grows. I estimate k by regressing observed variance on the player's

average inverse minutes for player-seasons in which a player plays more than 8 minutes in more than 20 games. The slope, $\hat{k}$, is approximately 650 (SE 21.7).⁷ Each player’s true variance is then his observed variance less this noise:

$$\hat{\sigma}_{\text{true},i}^2 = \max\left\{ \text{Var}_{\text{obs},i} - \hat{k} \mathbb{E}_i[1/m_{i,g}], 0 \right\}. \quad (2)$$

Hereafter, a player’s volatility refers to the square root of this adjusted variance.

Figure 1 demonstrates the importance of this correction. In the raw data, players who play less often look more volatile than those who play more minutes. If one had a prior belief that volatility was bad and steadier players should play more often, this would offer confirmation of that bias. After correcting for finite-sample noise, the relationship is roughly flat. If anything, the slight uptick in adjusted volatility suggests that the most important players are more volatile than those who are played sparingly.

Roughly 5% of player-seasons have observed variance below the estimated noise, given their minutes played. These players are so consistent that their game-over-game variation in performance is indistinguishable from the “bouncing ball” luck inherent to basketball. Figure 2 shows the spread of player-seasons from the sample period.

3. The Market Price of Reliability

3.1. Compensation and the Collective Bargaining Agreement

In the NBA, the rules governing on-court play and roster-building are negotiated between the league (representing the owners of the 30 franchise teams) and representatives for the players’ union (National Basketball Players Association or NBPA) as part of a collective bargaining agreement (CBA). Each CBA is an opportunity to substantially reshape the league landscape and the

⁷The slope $\hat{k}$ is identified from cross-player variation and inherits any correlation between volatility and playing time. Within-player and composition-based estimates of the same slope are larger and so would remove some volatility which persists within player across seasons. The appendix shows results for alternate values of $\hat{k}$ and provides a split-sample estimate to gauge the magnitude of measurement error.

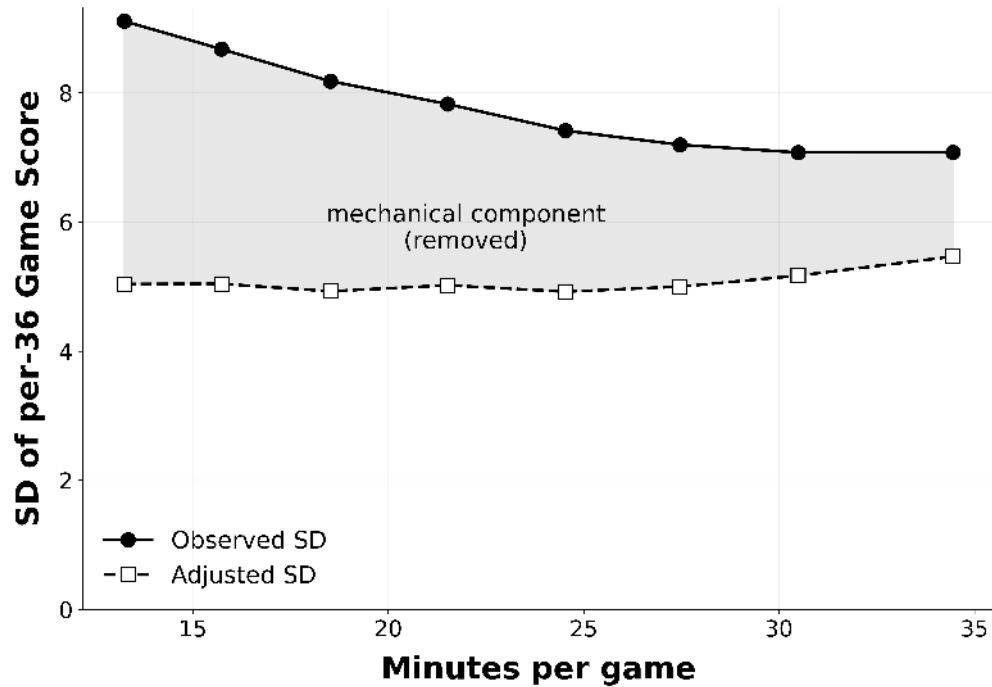


Figure 1: Observed and adjusted standard deviation of per-36 Game Score, by minutes per game. Points are bin means. The shaded region is the mechanical component removed by Equation (2). Chart shows player-seasons with at least twenty games of eight or more minutes, 2011–2023 ($n = 4,903$).

relationship between workers, fans, and teams. As such, the league’s history is typically broken up into eras defined by the prevailing CBA. This analysis is concerned with the “modern” NBA era, which began with the 2011 CBA and was extended through the 2023-2024 season. The data set, drawn from Basketball Reference, includes 346,785 player-games.

The most important and contentious feature of a CBA negotiation is financial: the share of league revenue to be paid in salaries, the salary cap which all teams will adhere to, and all accompanying compensation rules. This includes the value, length, and payment terms of player contracts. There are three contract types of interest which are available to players in free-agency (able to negotiate compensation with multiple teams). First is the minimum contract, which ensures that no player will receive less than \$1.1 million (in 2023-2024) per year. Veteran players earn larger minimum salaries, subsidized by the league: a player with 10+ years in the NBA by 2024 could not be paid less than \$3.2 million per season. The minimum contract is also the only contract

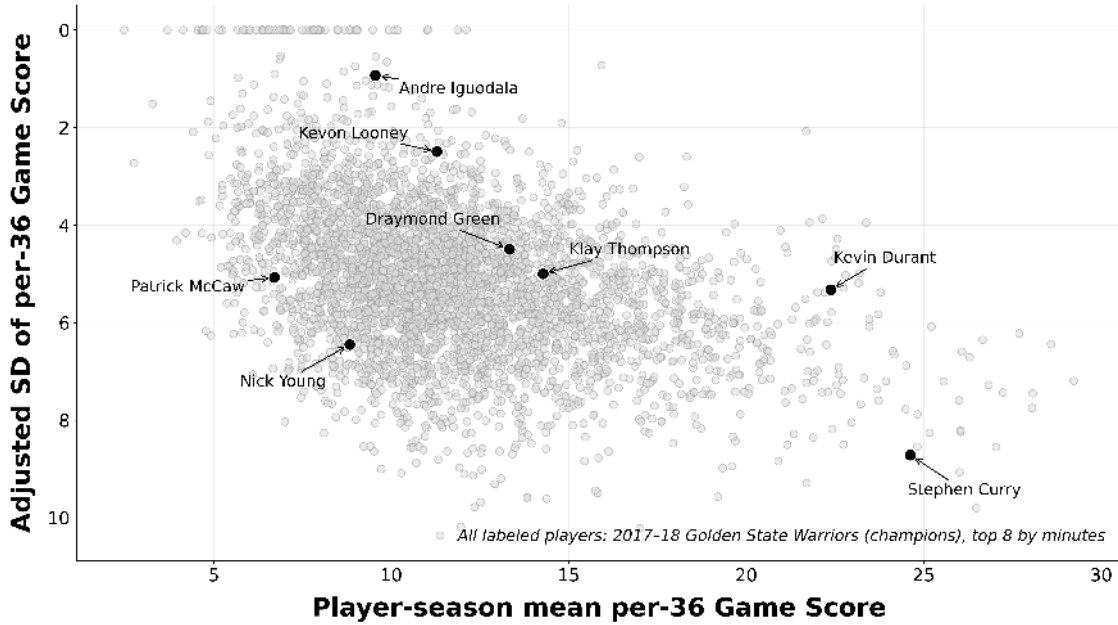


Figure 2: Scatter of player-seasons with 40+ games played from 2011-2023 ($N = 3,566$) by mean per-36 Game Score and adjusted volatility. Labeled points from the 2017-2018 Golden State Warriors.

freely available⁸ to teams whose payroll goes over the salary cap (~\$136m in 2023-2024) and is generally reserved for players who are at or near the replacement level or quality floor of the league. While many minimum players provide good value to their teams, they are generally nearer to retirement or unemployment than to superstardom.

A second contract type is the Maximum contract, which pays a percentage of the salary cap. The specific percentage – 25, 30, or 35% (the “supermax”) – depends on the player’s years in the NBA and awards won in seasons prior to signing a new contract. The Maximum contract is almost exclusively reserved for the best players in the league (or players that teams project to become superstars), but it substantially underpays them. In 2013-2014, for example, Blake Griffin was paid just over \$16 million, but generated on-court performance that would have been valued, on an uncapped free-agent market, at around \$34 million⁹.

Finally, players who are not offered a maximum contract but are too good to play on a

⁸There are various and limited exceptions to this rule, including the mid-level exception.

⁹Following FiveThirtyEight, I estimate a player’s open-market wage as the product of their wins (e.g., Win Shares) and the prevailing market price per win (approximately \$2.8 million in 2013–14).

veteran minimum negotiate wages in between the two extremes. The market for this “middle-tier” of free agents is, by no means unfettered or efficient. Teams are constrained from spending freely by the salary cap and their prior salary commitments. Rookie contracts are fixed by draft position and early-career players have their bargaining power reduced via “Restricted Free Agency” which gives their current team considerable leverage over the player and against other bidders. Players also value other compensation besides money. Some prefer to live in Los Angeles or to play with a superstar, beloved coach, close friend, or blood relative. Still, this is a substantial set of transactions in which the market values a player’s performance and the primary place in which a researcher can observe the wage impact of small changes in a player’s performance profile.

The NBA is often called a “superstar’s league” because team marketing, fan engagement, and competitive success tend to follow the best players. As such, teams are sophisticated in measuring and compensating top talent. A player whose expected performance justifies a maximum contract will likely have several suitors and the oddities of the CBA mean that the intensive margin of compensation with respect to performance is nonexistent. The same is true, albeit for opposite reasons, for players who are not good enough to earn above the minimum contract. If organizations value the distribution of a worker’s performance, it should show up in the middle tier of contracts alone.

3.2. Methods and Data

Across the data set, I observe 2,347 free-agent contract signing events and classify them according to the CBA’s wage scales. Of these, 201 are maximum contracts and 570 are minimum contracts (with both identified by the CBA’s published wage scales). The remaining 1,576 are middle tier, and are identified by the percentage change in annual value. Contractual raises cannot exceed 8%, so any large increases must be achieved in free agency. I also include salary decreases of the same magnitude.¹⁰ This approach excludes players who re-signed at essentially unchanged salaries. For clarity, these three tiers are subsequently subscripted as c_{cap} , c_{floor} and c_{mid} .

¹⁰I classify a signing as newly negotiated when its first-year salary differs from the player’s prior salary by more than 9% (the 8% cap plus a small buffer for proration and reporting error).

At each contract signing event, the primary dependent variable is the player's log-transformed first-year salary on the newly signed contract. The independent variable is the player's volatility (noise-adjusted Game Score SD) in a window of 1, 2, or 3 prior seasons. For each tier, I regress compensation on a measure of the player's prior-period performance volatility, controlling for prior-period mean performance and a battery of career-stage and role covariates (some shot profiles are inherently more volatile than others). The regression includes season fixed effects. Standard errors are clustered by player. Formally, for player i 's contract in time t :

$$y_{i,t} = \beta_{\sigma} Volatility_{i,t-1} + \beta_{\mu} Mean_{i,t-1} + Controls_{i,t} + \delta_t + \epsilon_{it} \quad (3)$$

Table 1: Variables and sample means by contract tier (signing events, 2011–2023)

Symbol	Definition	Mean by contract tier			
		Min	Mid	Max	All
y_{it}	First-year salary (\$M) of the newly signed contract; enters regressions in logs	1.30	7.08	29.04	7.56
$\text{Volatility}_{i,t-1}$	Noise-adjusted SD of per-36 Game Score over the evaluation window	4.73	5.08	5.69	5.10
$\text{Mean}_{i,t-1}$	Mean per-36 Game Score over the same window	9.74	11.41	15.81	11.67
age_{it}	Player age in signing year t	26.1	26.7	28.5	26.7
age_{it}^2	Squared age	—	—	—	—
$\text{tmin}_{i,t-1}$	Total minutes played, season $t - 1$; enters in logs	861	1,624	2,244	1,494
$\text{FGA}/\text{min}_{i,t-1}$	Field-goal attempts per minute, season $t - 1$	0.303	0.336	0.443	0.337
$\text{3PA}/\text{min}_{i,t-1}$	Three-point attempts per minute, season $t - 1$	0.101	0.105	0.127	0.106
$\text{FTA}/\text{min}_{i,t-1}$	Free-throw attempts per minute, season $t - 1$	0.073	0.086	0.133	0.087
δ_t	Signing-year fixed effects	—	—	—	—
N (signings)		570	1,576	201	2,347

The coefficients of interest are β_σ and β_μ . Both Table 2 and Table 3 report them in raw per-36 units: the change in log-wages associated with a one-point increase in per-36 mean Game Score or in the noise-adjusted per-36 standard deviation. Absent some bizarre circumstance, β_μ should always be positive, indicating that one additional point of Game Score is worth a higher salary. In raw units, the ratio $m_{mkt} = -\beta_\sigma/\beta_\mu$ represents the amount of mean Game Score a

team is implied to sacrifice, at a given wage point, in exchange for a one-point reduction in the noise-adjusted standard deviation of Game Score.

This analysis primarily examines the intensive margin, but volatility can also plausibly impact whether or not a player earns a contract of a given type or remains in the league at all. Using a multinomial logit model, I estimate the effect of increases to both mean performance and performance variance on the log-odds of a player's next free-agent contract falling into the middle or maximum tiers, rather than a minimum contract:

$$\ln \left[\frac{\Pr(\text{tier}_{i,t} = c)}{\Pr(\text{tier}_{i,t} = c_{\text{floor}})} \right] = \gamma_{\sigma,c} \text{Volatility}_{i,t-1} + \gamma_{\mu,c} \text{Mean}_{i,t-1} + \text{Controls}_{i,t} + \delta_{t,c}, \quad c \in \{c_{\text{mid}}, c_{\text{cap}}\} \quad (4)$$

I then use a simple linear probability model to test whether an at-risk player is re-employed. The sample keeps player-seasons of at least twenty games that are followed either by a new contract or by departure from the data set. Seasons whose continuation is contractually guaranteed are excluded, since those players mechanically cannot be forced to exit. This model uses the same set of controls and year fixed effects. Standard errors are clustered by player.

3.3. Results

For negotiated, middle-tier contracts, more volatile players earn less. Regression 1 evaluates log-wages on the prior season's game-over-game volatility without controlling for mean performance and recovers a positive coefficient, suggesting that more volatile players earn more than reliable players. As in other disciplines, the best NBA players are also players who play the most, do the most, and so have the largest spread on performance. However, the sign on β_{σ} reverses once the mean of the player is included in Regression 2. Over a single season, the penalty is statistically indistinguishable from zero (Regressions 2 and 3). It emerges, and grows monotonically, as the evaluation window is expanded to two and then three seasons (Regressions 4 and 5).¹¹

¹¹Sample sizes fall as the evaluation window widens because more prior-season data is required for the analysis.

A one-point increase in per-36 volatility (equivalent to +0.94 SD) measured over the prior three seasons is associated with a statistically significant 8.9% decrease in wages on the player's next free-agent contract ($p = .001$). When measured over two prior seasons, the penalty is 4.0% ($p = .06$). This trend reflects the difficulty of estimating a worker's variance compared to their mean (β_μ is precisely estimated in every evaluation window). As virtually all free-agent contracts are signed after 2+ (and usually 4+) years in the league, the coefficient in Regression 5 most accurately represents organizational attitudes towards worker volatility.

Volatility is persistent, rather than the artifact of a single season. When measured two seasons before the signing, or one season after it, a player's performance standard deviation predicts the wage penalty (Before: -0.036 , SE 0.011 ; and after: -0.040 , SE 0.012 , per point), and, obviously, post-signing volatility cannot have caused a contract that was already signed.

Table 2: Reliability premium on log(wage). Coefficients in raw per-36 Game-Score units; standard errors in parentheses, clustered by player. Columns (6) and (7) re-estimate the column (5) specification within the fixed tiers. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

	Negotiated Contracts					Fixed Contracts	
	(1)	(2)	(3)	(4)	(5)	(6) Min	(7) Max
β_σ (SD(GS))	+0.0318*** (0.0121)	-0.0049 (0.0125)	-0.0139 (0.0138)	-0.0412* (0.0223)	-0.0931*** (0.0290)	-0.0460 (0.0323)	-0.0141 (0.0307)
β_μ (Mean GS)	— —	+0.0594*** (0.0066)	+0.0519*** (0.0096)	+0.0646*** (0.0114)	+0.0910*** (0.0151)	+0.0240 (0.0163)	+0.0000 (0.0090)
age, age ²	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Log total minutes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Shot-attempt controls	No	No	Yes	Yes	Yes	Yes	Yes
Vol window (seasons)	1	1	1	2	3	3	3
Year FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes
N	1,176	1,176	1,176	925	677	77	123

The coefficients from Regression 5 are then used to generate estimates for the iso-value

slope m_{mkt} , or the inherent tradeoff between mean performance and volatility. All three evaluation windows have a positive value, suggesting that teams are willing to give up mean performance to get consistency at a given price point. However, the 95% confidence interval only excludes zero ([0.42, 1.63]) when teams have three seasons of prior performance to evaluate. Holding role, age, and shot profile constant, a value of 1.02 suggests that the market pays the same wage premium for a one-point increase in average performance as it does for a one-point reduction in volatility.

Table 3: Iso-value slope $m_{mkt} = -\beta_\sigma/\beta_\mu$, mid-tier signings. Raw Game-Score units; delta-method standard errors in parentheses; controls as in Table 2. * $p < 0.10$, *** $p < 0.01$.

Evaluation Window	β_μ	β_σ	m_{mkt}	95% CI	N
1 season	+0.0519*** (0.0096)	-0.0139 (0.0138)	+0.27 (0.26)	[-0.24, +0.78]	1,176
2 seasons	+0.0646*** (0.0114)	-0.0412* (0.0223)	+0.64 (0.34)	[-0.02, +1.30]	925
3 seasons	+0.0910*** (0.0151)	-0.0931*** (0.0290)	+1.02 (0.31)	[+0.42, +1.63]	677

Regressions 6 and 7 from Table 2 return the expected null results on the intensive margin. The minimum and maximum contracts offer pre-negotiated compensation, so a player who is already a max-worthy player does not earn more money by playing incrementally better nor is he punished for being more volatile. Table 4 reports the results of the multinomial logit and linear probability models which assess the relationship between extensive margin and performance volatility. Column 1 reports the difference in odds of signing a mid-tier contract versus a minimum tier contract. Column 2 compares maximum and mid-tier contracts. Column 3 reports the likelihood that a free-agent is re-employed in the subsequent season.

Unsurprisingly, the results for mean performance are positive and significant. A one-point increase in mean per-36 Game Score raises the odds of signing a middle-tier rather than a minimum contract by 21% and the odds of a maximum rather than a middle-tier contract by 30% (both

$p < .001$). However, the same increase in volatility has no significant effect.

Volatility likewise does not push players out of the league. Among at-risk player-seasons, re-employment rises by 1.9 percentage points per point of mean performance (SE 0.5) and by 1.6 percentage points per point of volatility (SE 0.8), suggesting that teams may be more likely to give one last chance to a volatile talent who briefly shows some flash of high-level performance.

Table 4: Volatility and the extensive margin. Columns (1)–(2): odds ratios per one-point increase from Equation (4), p -values in parentheses (MLE standard errors; column (2) is the maximum-versus-middle contrast, delta method). Column (3): linear probability of re-employment for free-agents, player-clustered standard errors in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

	(1) Mid vs. Min	(2) Max vs. Mid	(3) Re-employment
Mean (per-36 GS)	1.21*** ($< .001$)	1.30*** ($< .001$)	+0.019*** (0.005)
Volatility (SD per-36 GS)	1.00 (.976)	0.87 (.158)	+0.016** (0.008)
Log total minutes	9.37*** ($< .001$)	8.16*** ($< .001$)	+0.381*** (0.036)
Career stage (age, age ²)	Yes	Yes	Yes
Shot-attempt controls	Yes	Yes	Yes
Year FE	Yes	Yes	Yes
N	1,536	1,536	2,101

These results have two implications. First, they suggest that the results reported on the intensive margin are not the result of selection into the middle tier of contracts on the basis of a player’s performance spread. Second, they suggest that teams may evaluate performance sequentially. Initially, they assess how “good” a player is based on their mean performance. For players who are not worth a maximum contract but desirable enough to justify a mid-tier contract, teams then price volatility in conjunction with mean performance.

It is important to note that the observed relationship is not causal. Thus far, I have made

no effort to determine the compensation that a player would earn with their second-best offer or in some counterfactual in which they were more reliable but otherwise unchanged. Further, teams have more information about players than fans or researchers. Perhaps there are personality or environmental traits which make volatile players unattractive for reasons other than their on-court performance. Teams with detailed medical information and more insight into a player's precise tendencies may simply value dimensions of health, availability, or fit which are unavailable in the game/performance data.

While I remain unable to rule out the presence of privately known and unobserved (in this data set) heterogeneity, many of these confounders could plausibly bias the observed results towards zero. Information, in professional basketball, is a source of competitive advantage and so does not flow freely. This first set of results suggests that teams will pay a premium for reliability. Teams with information about some unattractive player characteristic will undervalue that player, relative to uninformed bidders. In all likelihood, the informed team will not sign the free-agent, leading players with hidden flaws to sort into uninformed teams. And it only takes one stupid, desperate, or risk-loving team to create a market for the headcase that 29 other franchises would prefer not to employ.

4. The Organizational Value of Reliability

Though these results suggest that teams value reliability in the free agents they sign, they say nothing about the normative value of that reliability. Nor does the question “should teams pay for steadiness” have an obvious answer. On one hand, professional basketball teams are generally “smart” and so one might assume that the decision to pay for something is rational and well-informed. On the other hand, research on organizations is replete with examples of mispriced attributes, biased preferences, and an over-adherence to received wisdom. One need not look further than professional sports, where baseball teams systematically overvalued batting average for generations (Hakes & Sauer 2006) and American football teams long resisted the statistically superior fourth down and two-point conversions (Hartley 2017; Romer 2006).

Even if we were convinced that streakiness is bad and consistency is good, it is possible that organizations have already evolved the tools and heuristics to manage it. In professional basketball, this might be the coach's scheme or playbook. In the broader set of organizations, this would be handled with routines and best practices. Of course, this may not even be necessary. After all, even compared to other professional sports, professional basketball allows for a large number of draws. A game has an average of 100 possessions (compared to 35-40 plate appearances per team in Major League Baseball) and a season has 82 games (compared to 17 in the National Football League and 38 in the English Premier League). And, unlike other sports, every player can contribute to every possession, so the law of large numbers has ample opportunity to insulate team outcomes from individual player volatility.

4.1. Methods and Data

Whether or not they are correct to price reliability, teams are undeniably strategic actors. They select their rosters and allocate playing time with clear objectives. In order to address the endogeneity of these decisions, I exploit the quasi-random timing of absences of key players. As a treatment, this has three useful features. First, while teams know that some players are more injury-prone than others, the precise timing of those injuries is difficult to predict. Further, teams use information about player injury risks strategically: fatigued players are given extra rest, prophylactic medical care, and tailored, state-of-the-art physical therapy, dietary advice, and equipment. Any injuries that result in spite of these measures are, I argue, an unpredictable residual. Finally, teams are heavily constrained in their mid-season roster choices, so absences almost always entail the reallocation of minutes to players who are already on the roster (and whose prior performance can be measured). In effect, each absence involves two pieces: the unexpected subtraction of a player and the promotion of their replacement. Both pieces are known and measured, so I compare the team's quality and volatility before and after the absence against team outcomes. The estimand is the effect of the subtraction event as teams experience it, including whatever strategic response the team can muster. Results are substantially unchanged when directly controlling for the quality and volatility of the

remaining roster.

I identify treatment events as player absences for which three conditions are met. First, the absent player must have averaged at least 15 minutes per game in the current season and played in each of his team's previous five games. This ensures that the absent players are active contributors (rather than experiments or fringe players) with sufficient performance history that I can examine pre-absence trends. Hereafter, I refer to these as "rotation players." Second, the player must play zero minutes for at least five games in a row.¹² This is a long enough spell that the team must genuinely reorganize its rotation in response. Finally, I keep only the first qualifying spell of each team-season so that one absence event (and the corresponding roster/rotation changes) do not contaminate subsequent events. We might imagine that a team responds to an injury to its second-best player by asking the best player to do more. This added workload may cause a subsequent injury and make it impossible to disentangle the effects of the two absences from each other and from the impact of the shift in workload. By using only the first qualifying absence of the season, I minimize this attribution problem.

For each event, the focal player's mean and volatility are measured over the prior season, using noise-adjusted, mean-centered Game Score and BPM. Game Score remains the primary trait of interest, but BPM is useful as it is a more robust measure of player mean quality in the long run. Players without a prior season (rookies, players who missed an entire season due to injury, players who were under contract but not good enough to play) are excluded. Across the 2011-2023 seasons, there are 268 qualifying events. Absent players average 27 minutes per game before their absence with a prior-season per-36 mean Game Score of 9.0 and volatility of 5.1. The median spell lasts 11 games.

I estimate the treatment effect of losing a player using a two-stage, staggered treatment difference-in-differences estimator (Butts & Gardner 2022; Gardner 2022). A conventional two-

¹²I do not use formal injury designations for two reasons. First, the data available is incomplete, with some verified injuries being reported as "Did Not Play - Coach's Decision" and others not reported at all. Second, there is nothing unique about injury as a reason for a player's absence compared to paternity leave, bereavement, or suspension for cause. For my purposes, these all result in the unexpected subtraction of a player from the team's roster for an indeterminate amount of time.

way fixed-effect design may produce biased estimates when treatment is staggered because some already-treated observations are used as controls for later cohorts. This estimator avoids this by ensuring that treated games are excluded from the counterfactual. I use Gardner (2022) instead of Callaway and Sant’Anna (2021) because the parameters of interest are coefficients on continuous moderators of the treatment effect. In the first stage (identical to Borusyak et al. 2024), I estimate team-season and game number fixed effects by regressing the result of team j ’s game g as follows:

$$y_{j,g} = \text{TeamSeasonFE}_j + \text{GameNumberFE}_g + \epsilon_{j,g} \quad (5)$$

on $\{(j, g) : \text{UntreatedGame}_{j,g} = 1\}$

This first stage uses games before and after the treatment period. I drop the last game before the absence because injuries which occur mid-game produce distorted statistics. I then estimate the gap between predicted and actual team results for all games:

$$\hat{u}_{j,g} = y_{j,g} - \widehat{\text{TeamSeasonFE}}_j - \widehat{\text{GameNumberFE}}_g \quad \text{for all } (j, g). \quad (6)$$

In the second stage, I regress this gap against post-period indicators for player mean and standard deviation:

$$\hat{u}_{j,g} = \tau \text{Post}_{j,g} + \beta_\mu \text{Post}_{j,g} \text{Mean}_i + \beta_\sigma \text{Post}_{j,g} \text{SD}_i + \epsilon_{j,g}. \quad (7)$$

I use two outcome variables. One is a simple result indicator (1 if win, 0 if loss). A second is the point differential, where positive numbers indicate the margin of victory and negative numbers the margin of defeat.

Following Gardner (2022), which treats the two stages as a single estimation problem, I compute standard errors that add the uncertainty inherited from the first stage to the second-stage variance. I cluster at the team-season level because first-stage estimation error is shared by every game within a team-season.

The coefficients of interest are τ , representing the average treatment effect on the treated (ATT), and β_μ and β_σ , which measure how the ATT varies with changes to the absent player’s

mean and volatility. Any claim to causality rests on a parallel-trends-style assumption. Here, that would mean that, absent the missing player, the treated team's results would have continued at the level implied by its own untreated games after adjusting for the usual rhythms of the NBA season. I discuss further below.

4.1.1. Treatment Timing

Figure 3 illustrates the construction of the pre-period, treatment period, and counterfactual samples. The first row illustrates a below-median length absence. The second shows a player whose absence is protracted. The game in which the player is injured or otherwise absented (region 2 in Figure 3) is not included in any samples because the player's statistics, in that game, are likely to be distorted by the mid-game injury.

I define the treatment-period as games 1-11 after the event or injury (region 3). This fixed window is designed to include the entire time period in which a team is adjusting to the absence of a key contributor. The median absence length is 11 games. I prefer a fixed post-treatment evaluation window length because a player's return from injury is partially a team choice. Some players are given long periods of time to recover to full health. Others are rushed back into action for a key game. Still others are traded and so never return to the team. Were the study to look at results only after the absent player returns, it would be capturing the effect of the team's timing decision.

The design estimates the team and time fixed effects from untreated periods of time (regions 1,4, and 5), which is defined to include games before the player is injured/absent and after the treatment window. Though this approach eases concerns about the endogeneity of a player's return, it is biased in multiple ways. For absences that are shorter than the 11 game evaluation window, the "treatment" includes games where the absent player has already returned (region 3a in Figure 3). Games the player misses beyond the 11 game window (region 4) enter the counterfactual pool, contaminating a baseline that is meant to represent the team with him available.

Because the treated event is the first qualifying absence of each team-season, no other qualifying absence is in progress when the window opens. However, when another rotation player

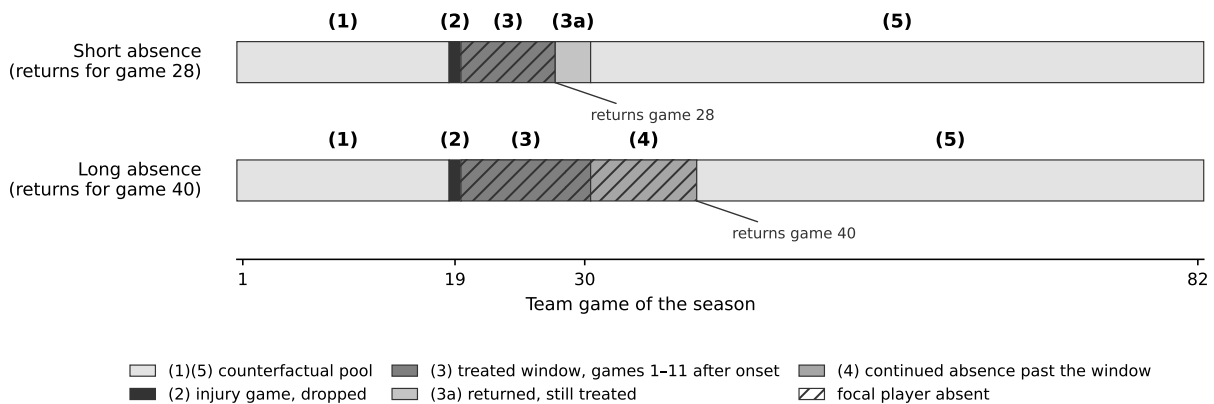


Figure 3: Illustration of treated and untreated periods for short and long injuries.

misses games before the first returns (roughly half of the events overlap with another absence), I treat it as part of the effect being measured because subsequent injuries could plausibly have been caused by overexertion in response to the first absence. Fortunately, the biases introduced by both short and long absences work against the reported ATT.¹³ Games in Region 3a (player is available but assumed to be absent) reduce the observed impact of the absence. Games in Region 4 (player is absent but assumed to be available) will lower the baseline and so reduce the observed difference between counterfactual and treatment periods. The same logic also explains why I do not control for other absences to key rotation players outside the treatment window (though tests of the effect of non-focal player absences are given in the appendix).

4.2. Results

By using only absences to rotation players as the treatment, this study virtually guarantees that the average event hurts the team. A team plays its best players more than its worst players. When a player who receives significant playing time is absent, the team can only allocate those minutes to players that are worse (else they would have been playing in lieu of the lost player) or to players who are better but playing at capacity (else the team would already have allocated more minutes to them). Table 5 reports the expected negative and significant coefficient on the post-period indicator

¹³Appendix includes estimates from alternative specifications.

for both outcome variables. Any absence costs the team, on average, about a point per game and just over two percentage points of win probability.

We would similarly expect a negative coefficient on β_μ . When a better player is absent, the team is hurt worse. Relative to the average absent rotation player, each additional point of prior-season Game Score costs the team about a sixth of a point of result margin and just over half of a percentage point of win probability. The positive sign on β_σ for all specifications suggests that an increase in the standard deviation of the absent player (holding mean performance constant) actually blunts the impact of their loss.

Table 5: Two-stage difference-in-differences estimates. Each row is a separate regression with different outcome variables. Two-step standard errors, clustered by team-season, in parentheses. $n \approx 21,500$ team-games; 268 treated cohorts. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

	<i>Game Score</i>			<i>Box Plus Minus</i>		
	τ	β_μ	β_σ	τ	β_μ	β_σ
Point differential	-0.974*** (0.337)	-0.169* (0.090)	+0.612*** (0.189)	-0.974*** (0.334)	-0.319** (0.136)	+0.380** (0.154)
Win rate	-0.022* (0.012)	-0.0052* (0.0031)	+0.0222*** (0.0072)	-0.022* (0.012)	-0.0088* (0.0046)	+0.0124** (0.0055)

To illustrate the magnitude of these results, consider four hypothetical players (Table 6). Player A is the average absent rotation player in the sample, with a per-36 mean of 9 and volatility of 5. Because the moderators are mean-centered, the cost of losing him is about one point of margin per game, or a quarter of a win over the median eleven-game absence. Player B is a near-star whose absence clearly hurts the team. Player C is the steady performer whose absence is greatly missed. Player D illustrates that sufficient volatility can completely offset the win-loss contributions of a league-average rotation player.

Table 6: Predicted cost of absence for four hypothetical players (Game Score coefficients, linear extrapolation).

	Mean (GS)	Volatility (GS SD)	Point differential per game	Losses (per-11 games)
A: average player	9	5	-1.0	+0.24
B: near-star	13	5	-1.7	+0.47
C: one point steadier	9	4	-1.6	+0.49
D: one point streakier	9	6	-0.4	0.00

4.2.1. Causality and Robustness

Figure 4 presents an event plot in which each bucket holds four games. In the four games before the absence and the first four afterwards, team point differential is statistically indistinguishable from zero (joint Wald $p = 0.49$ and $p = 0.16$ respectively). One possible explanation is that teams take immediate tactical and strategic steps to close the gap. However, the increased load on the remaining rotation (plus the opportunity for opposing teams to respond strategically) means that the relief is only temporary. In the subsequent 8 games, team performance falls significantly before rebounding in the final two buckets. The figure also splits the events into two cohorts above and below median volatility, suggesting that the ATT is largely driven by the absences of more reliable players.

The estimates in Table 5 support a causal story only if the games in the post-event evaluation window would otherwise have resembled the team’s season-long performance. More specifically, causality is threatened if there is something about the treatment window that impacts team performance, is correlated with absences, but does not result from player absences. This is different from other types of event studies which use the pre-treatment period to estimate the counterfactual and so must show that the post-treatment outcome was not already in progress when the event arrived.

I examine several plausible candidates. I first examine whether overexertion could be responsible. If absences (particularly injuries) follow stretches of unsustainable, high-effort play, the subsequent results would be poor with or without absences. However, recent form has no

consistent relationship with subsequent absences and the measured effects are unrelated to the team's pre-event performance (β_σ moves from 0.61 to 0.59 when an explicit pre-event-performance control is added). Likewise, there is no significant difference between treated and untreated games on location (home/away), opponent strength, or rest days between games. Finally, teams late in the season may change priorities (sometimes called tanking) from winning games to developing younger players or trading rotation players for other assets. The study design nearly rules this effect out by construction since only 5 of the 268 events begin at or after game 55.

Placebo tests using games before the event or after the absent player returns have negligible coefficients for margin of victory/defeat (pre-injury $\beta_\sigma = +0.08$, post-return $\beta_\sigma = +0.17$ for Game Score; +0.01 and +0.03 for BPM). Likewise, when the absent player's characteristics are replaced with those of a different, randomly selected player, estimates exceeding those in Table 5 are only reproduced in 20 out of 2,000 re-runs using BPM. For Game Score, it is 4 out of 2,000. Taken together, these robustness checks (reported in the appendix) suggest that whatever drives β_σ operates only while the player is not playing.

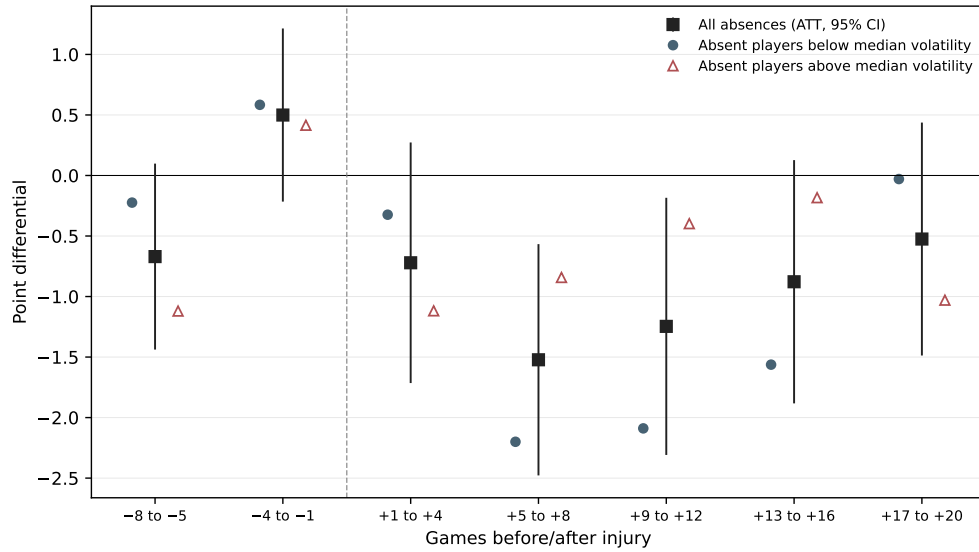


Figure 4: Team point differential around a rotation player's absence, in four-game buckets (the injury game, game 0, is omitted). 95% intervals from two-step standard errors, clustered by team-season.

4.2.2. On Steadiness and Superstars

The results presented in Table 5 are largely driven by players who excel on one or more dimensions. Figure 5 reports the binned ATT by quintile of prior season mean or volatility. The bins are balanced on playing time. Where previous plots use Game Score, I now shift to using Box Plus Minus (BPM) because it is a more sophisticated measure when making player-to-player comparisons. BPM's game values are estimated at the season level and are not interpreted here. However, the game-to-game dispersion is at least as reliable a measure of spread as Game Score's. Once more, negative values indicate the estimated harmful impact to team performance from losing a player with mean or standard deviation in a given quintile. The results for mean are intuitive. Losing below average players does not hurt the team nearly as much as losing a very good player (BPM above 4.0 is considered "All-Star Quality" per Myers (2020)).

Likewise, the loss of one of the most reliable players hurts team point differential. What is somewhat surprising is the fact that the magnitudes of the two effects are somewhat similar. Losing a star player costs the team 2.64 points per game. Losing a very reliable player costs 2.14 points.

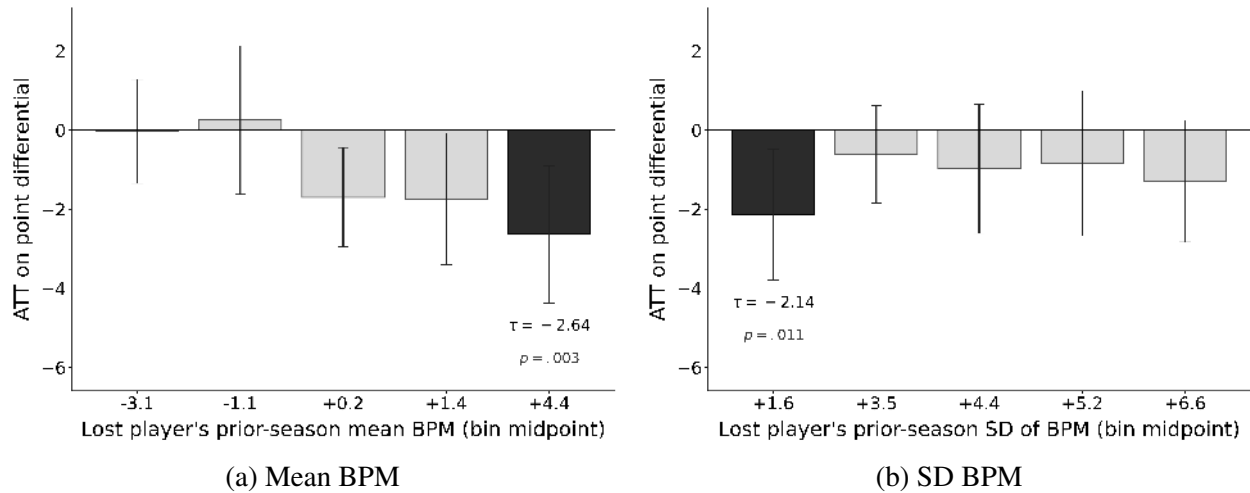


Figure 5: Post-injury ATT on point differential by minutes-balanced quintile of the absent player's mean or SD of prior-season BPM. Error bars are $\pm 1.96 \cdot$ bootstrap SE.

Interestingly, these appear to be almost completely different populations of players. Figure 6 plots the observed player absences and highlights the players in each of the top quintiles from

Figure 5. The overlap is tiny. There are only four players (out of 268) with observed absences who are both all-star quality and highly reliable. Of course, this is not to say that players cannot be both. Rather, the rare player who is a highly reliable superstar is also unlikely to get injured early in the season. Still, evidence suggests that Reliable (triangles) and High-Average (squares) are two different populations with roughly similar marginal replacement costs. Crucially, the former delivers 80% of the impact while being paid, on average, 40% as much as the latter group (Table 7).

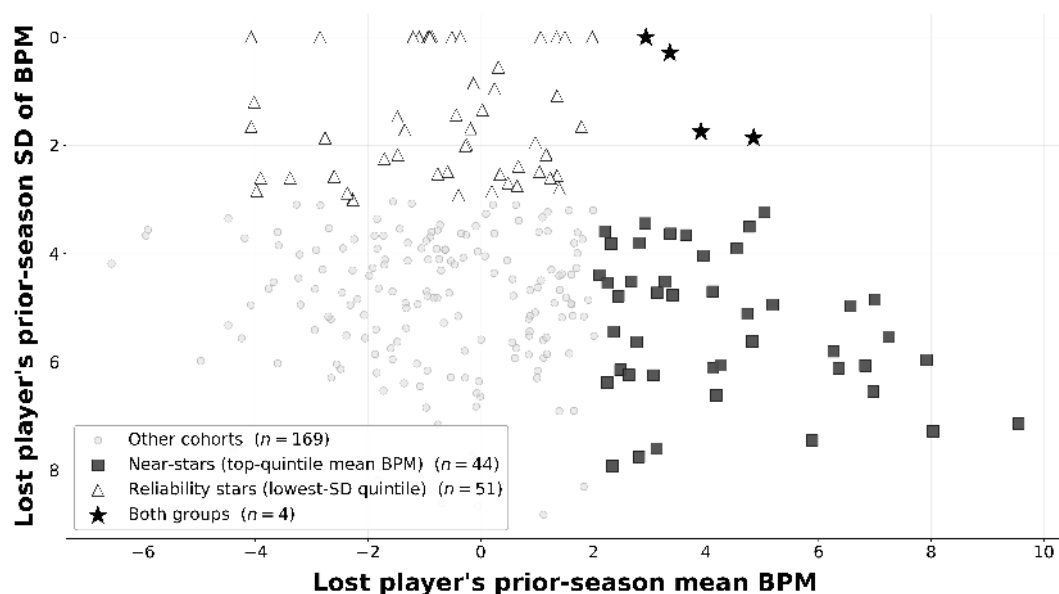


Figure 6: Absent or injured players by prior-season mean BPM and prior-season SD of BPM. Triangles represent players in the top quintile of Reliability. Squares are the top quintile of mean performance. Stars are in both categories.

Table 7: The price of losing a player: minutes-weighted ATT on point differential by quintile of the absent player’s prior-season BPM with salary and share of salary cap. Bootstrap standard errors in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

	ATT (pts/game)	Salary	% of cap
<i>Panel A: quintiles of mean BPM</i>			
Q1 (lowest mean)	−0.03 (0.67)	\$4.5M	5.8
Q2	+0.25 (0.96)	\$7.9M	9.1
Q3	−1.70*** (0.64)	\$10.0M	11.1
Q4	−1.74** (0.85)	\$12.1M	13.4
Q5 (near-stars)	−2.64*** (0.89)	\$18.7M	20.0
<i>Panel B: quintiles of SD of BPM</i>			
Q1 (most reliable)	−2.14** (0.84)	\$7.5M	9.0
Q2	−0.62 (0.63)	\$10.9M	12.8
Q3	−0.98 (0.83)	\$9.1M	11.3
Q4	−0.84 (0.94)	\$11.5M	12.1
Q5 (most volatile)	−1.29* (0.78)	\$12.4M	12.6

5. Discussion

The wage study, which shows that organizations do place a premium on reliable workers, has been done before with conflicting results. Barnes and Morgeson (2007) suggest that there is a negative association between performance variability and wages. Awtrey et al. (2021) find that organizations do not pay an appropriate premium for reliability. Neither analysis separates persistent, player-specific volatility from variation generated by playing time nor allows for the compensation nuances in the NBA CBA. After adjusting for these measurement and institutional constraints, I show a positive, significant relationship between a player’s game-over-game reliability and their negotiated compensation. The association also emerges more strongly as teams accumulate multiple seasons of performance information, consistent with the greater difficulty of estimating volatility than mean

performance.

The second analysis suggests that volatility is genuinely detrimental to organizational performance and outcomes. Even after the best efforts of coaches, managers, and teammates, who have more information about players' roles and tendencies and the ability to adjust schemes and rotations strategically, there is still a persistent and negative effect when a reliable cornerstone of the team is lost to injury. This validates the finding that teams prize reliability as a distinct player performance characteristic. The coefficients in Table 3 are consistent with those in Table 5. Organizations trade roughly one point of mean performance for a one-point reduction in Game Score standard deviation. They could reasonably value that same reduction at 3.6 points of mean production, though the ratio is imprecise (95% CI -0.5 to $+7.8$).

In the real world, superstars are rare. Not every company can have one, let alone an army of 10x engineers or A+ sales reps. And for companies with limited resources, top-tier human capital is often out of reach on the labor market. These results suggest that those organizations may find that it is better to build a steady and consistent, if unspectacular, workforce, rather than compete with the deep pockets of the world's largest and most prestigious firms. This is not to say that superstars are not important (most of the aforementioned highly reliable players share a roster with one or more max-contract superstars), but for companies who are constrained relative to their competitors in the market for talent, reliability has a quality all its own.

Missing, from this analysis, is more detail on the exact mechanism that drives the results. After all, I have shown that players who consistently score fewer points (or do fewer things) can be, on average, better for the team than those that score in bunches with fallow periods in between, which is, perhaps, counterintuitive. As this is a paper about organizations, performance, and strategy and not a paper about professional basketball, the specific basketball mechanisms are largely outside the scope of the project. However, I can note three mechanical features that generalize beyond professional basketball.

First, success and failure are binary and separated by a knife's edge. The first point that the opponent cannot match is extremely valuable (the team wins by 1) but the second point of

win-margin is worthless (a win by two points is the same as a win by one point).¹⁴ In effect, the team is forbidden from keeping any excess performance past some threshold. This is a common feature of organizational payoff functions and suggests areas where reliability or safety should be valued: Political parties need only win by one vote. Entrants need only move fast enough to erect entry barriers behind them. And startups need only enough revenue to pay the bills in order to live to fight another day. In environments with the opposite incentives (limited or fixed downside, lots of upside), we would expect to see a premium on volatility (venture capital investing, computational/algorithm races).

Second is the shape of worker performance itself. Game-level output is right-skewed for roughly 90% of the rotation players in professional basketball. A player's best decile of games averages 1.9 standard deviations above his own mean, while his worst decile averages only 1.5 below.¹⁵ Volatile players, then, do not trade good games for bad games, one for one. They are trading many games below average for a handful that are well above average, with team payoffs again, limited by the nature of binary win-loss outcome.

Finally, as is true in many organizations, basketball is a team sport in which a player's contributions affect both the scoreboard and the contributions of their peers. A reliable teammate, in basketball, helps the offensive and defensive schemes function even when other players are out of position. They provide insurance that covers for mistakes or allows others to take risks without fear of outsized negative consequences. In settings where performance is entirely independent, we should see a weaker preference for reliability (as in professional baseball, which values power hitters highly, even if they strike out often). And where teamwork matters, it is worth paying a premium for consistency.

¹⁴Though professional basketball has flirted with rule changes which would modify this, including the Elam ending's target-score finish.

¹⁵Player-seasons with at least forty appearances and fifteen minutes per game, 2011–2023.

6. Conclusion

This paper suggests that organizations do price worker volatility consistent with its impact on organizational outcomes. Where the CBA allows prices to move, professional basketball teams pay a premium for player consistency. As evidence of a player's volatility accumulates, teams are more likely to reward consistent players and pay less for the same mean performance from volatile players. When an event removes a worker, the cost of his absence rises with his steadiness (holding average performance fixed) and is not inconsistent with the wage the market pays.

In strategic factor markets, firms can capture value when they have more accurate expectations than competitors about the future value of a resource or the evolution of the world state (Barney 1986), which makes the generation, gathering, and interpretation of information into important capabilities for sustained competitive success (Makadok & Barney 2001; Ryall & Sorenson 2025). Though people are the eponymous human resource, strategic human capital would benefit from developing on these well-understood ideas. After all, worker reliability is important to organizational success, difficult to know ex-ante, costly to estimate, and slow to be reflected in labor-market prices. Organizations that measure it more accurately may have a substantial and sustained advantage over their competition. Substantial because the production value of reliability is available immediately, but the labor market responds more slowly.

Obviously, organizations that excel at identifying or manufacturing reliability can sustain this advantage over an extended period of time. However, there is also evidence that workers' reliability is not fixed. When an individual's penchant for good or bad days evolves, it lengthens the window of time in which their productivity will be mispriced. In effect, the labor market can be precise about a worker's volatility or current, but not both. An organization that observes its workers daily and so can more quickly and accurately gauge the second moment of the performance distribution has no such tradeoff. As the worker evolves, the organization's advantage is renewed.

Finally, I return to the question of why volatility is so conspicuously absent from the literature on human capital strategy and organizational design. After all practitioners and researchers are

aware of the Peter Principle. Why hasn't research extended the concept further?

This paper shows that at least one type of organization does price worker volatility where it is observed, so it's difficult to argue that decision-makers and researchers have a blind spot or bias. And even in a setting with ample strategic behavior and substantial scale (100 possessions per game, 82 games per season, each with 10 workers participating), the normative effect remains. Losing someone consistent produces an outsized and negative effect on team performance. This is not to say that buffering or formal routines and structures are irrelevant, but it does suggest that they are incomplete and costly enough to present organizations with a dilemma. Likewise, the law of large numbers cannot fully average away outlier performances in a competitive setting with fine margins.

What remains is measurement. While there is substantial evidence that volatility is a semi-stable worker characteristic, there is widespread disagreement about the shape of the distribution from which performance is drawn (Beck et al. 2014; O'Boyle Jr. & Aguinis 2012). Without knowing whether a worker's performance is normal, log-normal, or heavy-tailed, how much data must an employer collect before confidently calling that worker reliable or volatile? And for researchers, attempting to simultaneously estimate both moments of each worker's performance distribution is likely to strain most field and lab studies. Even in this setting, more than half of observed variance is mechanical noise and the wage penalty is only visible in the data over multi-season evaluation windows. But, where measurement is available, rational and strategic behavior follows.

We should start there. Research on human resources should either have a credible reason for reducing workers to the mean or take steps to ensure that interventions which raise the average do not materially and adversely impact the distribution of worker performance. Measurement, in turn, gives way to management, which is important because workers do not produce averages. They deliver noisy, messy sequences of good and bad days, and the organization gets to experience them all.

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Appendix to *Playing the Spread: Worker Reliability and Organizational Performance*

APPENDIX A: ESTIMATING PLAYER VOLATILITY AS A CHARACTERISTIC

A player's observable game-over-game volatility is closely related to the amount that they play. Players who see the court less in each game have a higher game-over-game variance, so an accurate estimate of player volatility depends on being able to separate noise associated with short (or long) playing stints from persistent, player-specific volatility. The parameter k determines how much to weight playing time in the variance decomposition. In the main analysis, I estimate $k = 650$ from 4,903 player-seasons in which a player plays 20 or more games with 8 or more minutes. When applied to all player-seasons (5,431) of at least 10 games and 8 minutes played, this removes a median 53% share of observed variance and suggests that roughly 5% of player-seasons are so consistent in their performance that we cannot separate individual volatility from chance. The median player-season standard deviation of per-36 Game Score falls from 7.58 to 5.17.

Table A.1 shows a series of additional values for k and their impact on the sample statistics. Larger values of k result when estimating the impact of differences in playing time using only within-player minutes variation. Subsequent rows further attempt to remove sources of noise for a given player by controlling for the number and type of shots they take (the largest source of noise in the Game Score statistic). The last row abandons regression entirely and instead estimates k_i for each individual player from his own mix of shots (twos, threes, free-throws) and same-season shooting performance.

Table A.1: Alternate noise slopes k and their consequences for the volatility measure.

k	Estimated from	Share of variance removed	Players with zero volatility	Median Adjusted SD	Corr(SD_t, SD_{t+1})
0	no correction (raw SD)	0%	0.0%	7.58	0.35
650	cross-player regression (used in the paper)	53%	5.1%	5.17	0.33
999	within-player minutes variation	81%	28.3%	3.30	0.39
1,051	usage-adjusted, at mean shot volume	85%	32.7%	2.92	0.39
1,066	shot-mix composition, median player	86%	34.0%	2.81	0.39
k_i	shot-mix composition, own slope per player	86%	26.2%	2.79	0.05

Table A.2 shows how variation in k impacts the main results shown in Table 2. That is: middle-tier free-agent salaries with three years of prior playing history. The coefficient β_σ shrinks, as expected, when more variation in variance is removed from the sample. However, it remains significant for all except the player-specific $k = k_i$.

Table A.2: Three-season wage coefficients under each candidate noise slope k . $N = 677$. Standard Errors clustered by player.

k	β_μ	(SE)	β_σ	(SE)
0	+0.0877***	(0.0145)	-0.1594***	(0.0346)
650	+0.0910***	(0.0151)	-0.0931***	(0.0290)
999	+0.0890***	(0.0152)	-0.0499**	(0.0212)
1,051	+0.0879***	(0.0152)	-0.0422**	(0.0206)
1,066	+0.0877***	(0.0152)	-0.0405**	(0.0205)
player-specific	+0.0807***	(0.0145)	-0.0241	(0.0208)

As an additional check to ensure that the adjusted standard deviation reflects a worker-specific feature, I split each player-season into two halves (odd and even games) and compute the adjusted volatility separately for each half. Being drawn from a single player, the standard deviation in each half-season is a noisy estimate of the same characteristic. I then use one half as an instrument for the other. The wage regression itself is unchanged. Because the two halves share no games, this specification removes independent, game-over-game luck. Table A.3 compares the coefficients found in Table 2 (the OLS column) to the corresponding IV estimates.

On a single year of prior performance, the instrument is too weak to be useful. With two seasons of data, the magnitude of the IV estimate is comparable to the OLS estimate, but not statistically significant. However, the three-season evaluation window is a strong instrument ($F = 30$) and shows a larger (and still significant) effect compared to the OLS estimates. One odd feature of this approach is that teams observe (and choose wages) based on the full sample of games. So if they are sophisticated in attempting to parse out a worker's volatility, the IV coefficients are more representative than the OLS coefficients. If, instead, they are mostly reacting to observable behavior with a basic correction for minutes played, the true value is likely closer to the OLS estimates.

Table A.3: Split-sample instrumental-variable estimates of the wage response to volatility, on the estimation sample of Table 2. Coefficients are per SD point of volatility. Bracketed estimates have a first stage below the conventional threshold ($F < 10$) so standard errors are unreliable and no significance is assigned.

Window	OLS	(SE)	IV	(SE)	F	Split-half r	N
1 season	-0.0139	(0.0138)	[+0.0600]	[(0.1161)]	8	0.33	1,176
2 seasons	-0.0412*	(0.0223)	-0.0356	(0.0844)	27	0.45	925
3 seasons	-0.0931***	(0.0290)	-0.1554**	(0.0771)	30	0.51	677

Table A.4 tests the sensitivity of the results in Table 5 to changes in k . As before, the coefficient on β_σ decreases as more variance is pruned away but the estimates remain significant and material. Note that because the moderator's scale shrinks as k grows, the second pair of columns restates the coefficient per one standard deviation. With this normalization, the coefficients are largely invariant to the choice of k .

Table A.4: Estimates from the difference-in-differences design using injuries as treatment and volatility as a moderator (β_σ) for each alternate noise slope k . Moderators are measured in Game Score and the outcome variable is point differential. $N = 268$.

k	β_σ (per point)	(SE)	β_σ (per SD)	(SE)
0	+0.6705**	(0.2689)	+0.739**	(0.296)
650	+0.6122***	(0.1888)	+0.949***	(0.293)
999	+0.4570***	(0.1502)	+0.960***	(0.315)
1,051	+0.4606***	(0.1522)	+0.983***	(0.325)
1,066	+0.4627***	(0.1520)	+0.990***	(0.325)
player-specific	+0.3427**	(0.1649)	+0.576**	(0.277)

The choice of k is, ultimately, a judgment call. I have elected to estimate k from all players because it is intended to remove only the variation associated with minutes played. The objective is to estimate the persistent, individual *characteristic* and I believe it is safer (and more informative) to avoid controlling too much for observable differences in players. After all, those

differences are what the market prices and the organization derives value from.

A.1 Volatility as a Stable Trait

As shown in Table A.1, the season-over-season correlation in player-specific volatility is 0.33.

This analysis shows how consistent this characteristic is over 2 and 3 year spans (Rows 2 and 3) and between odd and even games (Row 4).

Table A.5: Within-career drift of the player’s volatility trait. Cells are correlations across the measure-construction frame of Table A.1 (5,431 player-seasons, 2011–2023). Row (1) for volatility matches that table’s final column at $k = 650$. Volatility is the noise-adjusted standard deviation of per-36 Game Score; the mean is per-36 Game Score.

	Volatility	Mean performance
(1) Correlation year-over-year	0.33	0.80
(2) Correlation, two years apart	0.29	0.74
(3) Correlation, three years apart	0.27	0.68
(4) Correlation, odd and even games	0.41	0.88

APPENDIX B: SELECTION AND SURVIVAL

More performance history is useful in estimating a player’s personal volatility. However, many players play fewer than four seasons (focal season plus three prior seasons as used in Table 2) and are subsequently not offered a contract. This is partially addressed by the analysis of the extensive margin, but as a robustness check, it is useful to ensure that the significant coefficient on β_σ with three prior years of data is not driven by selection.

I re-estimate the wage regressions using only the 677 free-agent signings for which there is a full three years of prior performance data in the sample. As in the manuscript’s main specification, a single-season of performance is insufficient. As more data becomes available, the coefficient becomes larger and more significant, suggesting that the set of players who survive long enough to be evaluated on volatility is not driving the observed results.

Table B.1: The reliability premium at one, two, and three seasons of prior performance. All-signings columns reproduce Table 2, columns (3)–(5), whose samples shrink as the evaluation window lengthens. Common-sample columns hold the sample fixed at the 677 signings with three seasons of prior performance. Coefficients on volatility in raw per-36 Game-Score units. Standard errors in parentheses, clustered by player. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Prior Performance Data						
Sample	1 season		2 seasons		3 seasons	
	All	Common	All	Common	All	Common
β_{σ} (SD(GS))	−0.0139	−0.0188	−0.0412*	−0.0620**	−0.0931***	−0.0931***
	(0.0138)	(0.0194)	(0.0223)	(0.0268)	(0.0290)	(0.0290)
Signings	1,176	677	925	677	677	677

APPENDIX C: ROBUSTNESS OF THE UNTREATED GAME DEFINITION

The manuscript uses a fixed, 11-game window post-injury as the treatment period in order to minimize the impact of strategic roster management. Teams have considerable discretion in when they bring players back from injury. An organization which desperately needs to win games may return a player to the lineup as soon as possible. One without that imperative may wait until they are fully healthy. However, the use of a fixed window allows untreated games to intrude into the treated window and treated games to fall outside the window. I have argued that both are likely to bias results to zero. Table C.1 replicates the results shown in Table 5 using a variety of different definitions of “untreated” games. Specification (1) is the standard, 11-game treatment window in which all games before the injury and starting 12 games afterwards are used along with never-treated games from other teams to construct the baseline. Specification (2) removes all consecutive games in which the injured player is absent and (3) further removes all games in which any important player is in the midst of a 5+ game absence. Finally, Specification (4) estimates the untreated level only from pre-treatment games and never-treated team-seasons.

As the criteria for untreated games becomes more restrictive, the coefficient for the ATT (that is, the average cost of losing a player) grows substantially. Meanwhile, the coefficient on

volatility (β_σ) remains positive across all specifications. Though more restrictive definitions lower statistical power (Specification 4's untreated pool contains less than one third of the games used for Specification 1), β_σ for Game Score only becomes statistically insignificant in the win rate panel when using the most restrictive Specification 4. The consistency of these estimates suggests that the results reported in the manuscript are not particularly sensitive to the games which are used to construct the untreated baseline.

Table C.1: Alternate definitions of untreated games. The four specifications use 18,608, 16,015, 8,774, and 5,458 games, respectively. Two-step standard errors, clustered by team-season, in parentheses; 268 events in every row. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

	<i>Game Score</i>			<i>Box Plus Minus</i>		
	τ	β_{μ}	β_{σ}	τ	β_{μ}	β_{σ}
<i>Point differential</i>						
(1) All games outside the treatment window (main text)	-0.974*** (0.337)	-0.169* (0.090)	+0.612*** (0.189)	-0.974*** (0.334)	-0.319** (0.136)	+0.380** (0.154)
(2) Also excluding the rest of the injured player's absence	-1.015*** (0.350)	-0.224*** (0.082)	+0.577*** (0.191)	-1.015*** (0.346)	-0.429*** (0.119)	+0.315** (0.157)
(3) Also excluding all qualifying absences	-1.416*** (0.392)	-0.290*** (0.092)	+0.633*** (0.216)	-1.416*** (0.390)	-0.483*** (0.137)	+0.299* (0.174)
(4) Pre-injury and never-treated games only	-1.446*** (0.498)	-0.178* (0.099)	+0.402* (0.243)	-1.446*** (0.495)	-0.374*** (0.145)	+0.221 (0.190)
<i>Win rate</i>						
(1) All games outside the treatment window (main text)	-0.022* (0.012)	-0.0052* (0.0031)	+0.0222*** (0.0072)	-0.022* (0.012)	-0.0088* (0.0046)	+0.0124** (0.0055)
(2) Also excluding the rest of the injured player's absence	-0.024* (0.012)	-0.0064** (0.0030)	+0.0222*** (0.0074)	-0.024** (0.012)	-0.0113*** (0.0044)	+0.0104* (0.0057)
(3) Also excluding all qualifying absences	-0.037*** (0.014)	-0.0075** (0.0034)	+0.0170** (0.0084)	-0.037*** (0.014)	-0.0130*** (0.0050)	+0.0063 (0.0063)
(4) Pre-injury and never-treated games only	-0.038** (0.017)	-0.0050 (0.0036)	+0.0123 (0.0091)	-0.038** (0.017)	-0.0116** (0.0053)	+0.0062 (0.0068)

APPENDIX D: ABSENCE TIMING

D.1 Strength of Schedule and Tanking

In the injury-based event study, the game-of-season and team-season fixed effects capture much of the temporal variation in the sample. However, it is possible that absences correspond to particularly challenging games. Because all wins count the same, a team might choose to rest players who are carrying nagging injuries (or are simply fatigued) during a stretch of difficult games. Though this is unlikely given that 5+ games would be an unusually long absence for load management, I additionally test whether absences (really, the 11-game windows following an absence) correspond to more challenging stretches of the schedule.

I estimate a team's likelihood of winning each game before the season begins using both teams' prior-season points for and against and an indicator for game location (home or away). I then combine this estimated win probability with the game's actual result to estimate each team's win rate in excess of expectation. This excess win rate is then used as the dependent variable in the same regressions shown in Table 5. As an alternate test, I re-estimate the main results without the five events that occurred late in the season, which are the most likely candidates for tanking (losing games in order to gain benefits in subsequent seasons).

The results in Table D.1 closely match those for Win Rate in the manuscript results table. Additionally, there is no material relationship between schedule difficulty and injury timing. Treated games are, on average, 0.1 percentage points more winnable than untreated games (SE = 0.3 percentage points), which is negligible.

Table D.1: The Table 5 specification re-estimated with win rate in excess of expectation as the outcome variable (first row) and dropping late-season events (second row). Two-step standard errors, clustered by team-season, in parentheses. 268 events in the first row and 263 in the second. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

	<i>Game Score</i>			<i>Box Plus Minus</i>		
	τ	β_μ	β_σ	τ	β_μ	β_σ
Win rate in excess of expectation	-0.023** (0.012)	-0.0053* (0.0030)	+0.0208*** (0.0070)	-0.023** (0.012)	-0.0093** (0.0045)	+0.0123** (0.0053)
Win rate, excluding events at game 55 or later	-0.022* (0.012)	-0.0044 (0.0031)	+0.0224*** (0.0072)	-0.022* (0.012)	-0.0079* (0.0047)	+0.0130** (0.0055)

D.2 Reversion to the mean

I further argue that injury events are quasi-random. Even knowing that a player is injury prone (or likely to be absent for other reasons, including paternity leave or suspension), teams cannot predict the exact point in time at which they are likely to become injured in advance. However, injuries can be caused by overexertion and absences can be the result of strategic responses to prior trends. Overexertion can result in unsustainable positive results before the injury. So what if the observed treatment results are, in fact, regression to the mean, helped along by a key absence? Alternatively, we could also imagine that injuries happen when teams are already trending downwards, either because a player is carrying some small injury which inhibits their performance or due to other, non-injury related factors.

Table D.2 shows the results of a linear probability model in which the team's form (average point differential) in the prior 3, 5, and 10 games predicts an injury event. Positive values mean that prior over-performance predicts an absence and is consistent with the idea that overexertion is partially to blame for the reported results.

The analysis is inconclusive. The prior three games have a positive but insignificant coefficient. When the window expands to five games, the coefficient becomes significant but

remains small: a five game period of over-performance increases the likelihood of absence by roughly 0.11 percentage points per added point of form. However, when the window is further expanded to 10 games, the sign flips and is, again, insignificant. It is tenuous to suggest that a positive performance driven by effort alone has a delayed effect on the player's injury likelihood, but we could contrive a story around elevated effort with diminishing returns over time.

Table D.2: Linear probability models of the start of a team's first qualifying absence on the team's trailing average point differential. Each entry is denominated in percentage points per point of form. Includes team-season and game-number fixed effects. Standard errors, clustered by team-season, in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Trailing games	Absence probability per point of form	(SE)	Team-games	Absences
3	+0.026	(0.046)	4,886	268
5	+0.113*	(0.066)	4,332	252
10	-0.089	(0.113)	3,167	169

I test the reversion story by re-running the main specification from Table 5 with the explicit inclusion of controls for pre-event form. The results remain consistent and significant.

Table D.3: The Table 5 point-differential specification re-estimated with pre-event team form added as a moderator. 268 events. Two-step standard errors, clustered by team-season, in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

	<i>Game Score</i>			<i>Box Plus Minus</i>		
	τ	β_μ	β_σ	τ	β_μ	β_σ
Point differential, pre-window form	-0.974***	-0.170*	+0.590***	-0.974***	-0.324**	+0.356**
added as a moderator	(0.340)	(0.089)	(0.188)	(0.337)	(0.135)	(0.151)

APPENDIX E: ROSTER AND REPLACEMENT QUALITY

Each absence or treatment is moderated by the worker's mean and volatility. However, each absent player is replaced and their minutes are allocated to other players on the roster. Because

mid-season roster changes are heavily constrained, this nearly always means that a player who has not previously played will enter the rotation, minutes will be reallocated to other players who will play more than previously, or some combination of the two. This is a strategic choice made with full information, so it is natural to question how much of the observed effect is a characteristic of the replacements as opposed to the absence itself.

I estimate this by measuring each team's per-team-minute mean and SD for the prior 10 games. I then measure the same value for games within the treatment window and use the difference between the two as a moderator in the main difference-in-differences design.

Table E.1 compares these analyses for point differential (row 2) and win rate (row 4) to the main specification from Table 5 (rows 1 and 3). The differences are negligible, suggesting that the treatment effect is tied to the player who is absent, rather than the players who replace them.

This is to be expected. Teams already play their "best" players (however that is defined) more than others. A replacement is, by definition, more limited in their ability to affect the team's outcomes. Moreover, because teams are all operating with similar roster-building constraints, the quality of replacement players is likely to be very similar. While there is considerable variation in the quality of each team's best player, the 11th best player on every team is likely to be, at best, slightly below average.

Table E.1: Replacement-player controls. Rows (1) and (3) reproduce the Table 5 specifications; rows (2) and (4) add the realized change in the remaining rotation's minutes-weighted mean and volatility as moderators. 268 absence events in every row. Two-step standard errors, clustered by team-season, in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

	<i>Game Score</i>			<i>Box Plus Minus</i>		
	τ	β_μ	β_σ	τ	β_μ	β_σ
<i>Point differential</i>						
(1) Main specification	-0.974*** (0.337)	-0.169* (0.090)	+0.612*** (0.189)	-0.974*** (0.334)	-0.319** (0.136)	+0.380** (0.154)
(2) Replacement change added as moderators	-0.974*** (0.337)	-0.115 (0.106)	+0.590*** (0.206)	-0.974*** (0.334)	-0.258* (0.141)	+0.376** (0.157)
<i>Win rate</i>						
(3) Main specification	-0.022* (0.012)	-0.0052* (0.0031)	+0.0222*** (0.0072)	-0.022* (0.012)	-0.0088* (0.0046)	+0.0124** (0.0055)
(4) Replacement change added as moderators	-0.022* (0.012)	-0.0023 (0.0036)	+0.0255*** (0.0079)	-0.022* (0.012)	-0.0053 (0.0046)	+0.0131** (0.0058)

APPENDIX F: PLACEBO TESTS AND PERMUTATIONS

As a test to confirm that the observed results are attributable to the absence event, I run two placebo tests. For each, I shift the "treatment period" to periods when the player was available. The backward placebo treats the 10 games before the injury as the post-treatment period. The forward placebo treats the 10 games after the true window as the new post-treatment period. The latter has fewer events because it only includes absences with fewer than 10 games which are not impacted by other events. For both placebo tests, the treatment effect (τ) and coefficient on player volatility (β_σ) are insignificant and comparatively small in magnitude as shown in Table F.1.

Table F.1: Placebo windows. Rows (1) and (4) reproduce the Table 5 results (268 events, two-step standard errors clustered by team-season). Rows (2) and (5) are backwards placebos (268 events). Rows (3) and (6) are forwards placebos (116 events). Placebo rows are estimated at the event level with robust standard errors. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

	<i>Game Score</i>			<i>Box Plus Minus</i>		
	τ	β_μ	β_σ	τ	β_μ	β_σ
<i>Point differential</i>						
(1) Main specification	-0.974*** (0.337)	-0.169* (0.090)	+0.612*** (0.189)	-0.974*** (0.334)	-0.319** (0.136)	+0.380** (0.154)
(2) Pre-injury window (ten games before the injury)	-0.098 (0.287)	+0.067 (0.077)	+0.077 (0.202)	-0.098 (0.287)	+0.127 (0.112)	+0.012 (0.156)
(3) Post-return window (ten games after the window)	-0.236 (0.552)	+0.112 (0.145)	+0.172 (0.377)	-0.236 (0.550)	+0.330 (0.221)	+0.035 (0.336)
<i>Win rate</i>						
(4) Main specification	-0.022* (0.012)	-0.0052* (0.0031)	+0.0222*** (0.0072)	-0.022* (0.012)	-0.0088* (0.0046)	+0.0124** (0.0055)
(5) Pre-injury window (ten games before the injury)	+0.002 (0.009)	+0.0021 (0.0024)	+0.0038 (0.0068)	+0.002 (0.009)	+0.0060* (0.0035)	-0.0010 (0.0051)
(6) Post-return window (ten games after the window)	-0.006 (0.019)	+0.0016 (0.0050)	+0.0092 (0.0147)	-0.006 (0.019)	+0.0047 (0.0070)	+0.0057 (0.0121)

Further, I use a permutation test to ensure that the observed results are not driven by a handful of outliers. Using the same 268 absences, I first randomly assign players (and their mean and volatility) to absences (Permutation 1). I then retain mean and assign a randomly chosen volatility from within the same strata of mean performance to each event (Permutation 2). This random assignment is performed 2,000 times. Table F.2 shows the frequency with which each permutation produces results with at least as large a magnitude as β_σ .

Table F.2: Permutation tests. Each count is the number of 2,000 draws in which permutations produce $|\beta_\sigma|$ at least as large as observed.

	Observed β_σ	(1) Mean and volatility reshuffled together	(2) Volatility reshuffled within mean quintiles
Point differential, Game Score	+0.612	4	3
Point differential, Box Plus Minus	+0.380	20	21
Win rate, Game Score	+0.0222	5	3
Win rate, Box Plus Minus	+0.0124	53	54

APPENDIX G: TREATMENT WINDOW ROBUSTNESS

When estimating the effect of an injury on team performance, I elected to use absences of at least five games and a treatment window equal to the median absence (11 games). For completeness, I estimate the treatment and moderation effects on alternate choices for both in Table G.1. Rows (1) and (6) reproduce the Table 5 results (268 events). Rows (2) and (7) count as treated only the games the player misses within the eleven-game window. Rows (3) and (8) count every game of the absence instead of some fixed treatment window length (268 events each). Rows (4) and (9) raise the qualifying-absence minimum to seven games (264 events). Rows (5) and (10) raise it to ten games (239 events).

When the treatment window is tightened to include only games missed within the 11-game treatment window, the coefficient on Volatility increases in magnitude. The only alternate specification that is materially different from the main results in the manuscript occurs when the treatment window is exactly the games missed, regardless of absence length. Though the sign is consistent for Game Score, the magnitude is smaller. For Box Plus Minus, the sign reverses. I attribute this difference to adaptation on the part of the teams who have lost players. Teams whose players are absent for an extended period of time adjust. Players take on new roles. Playbooks are modified. Strategies and expectations are revised. These adaptations are necessitated by the absence, but they are not the same as the effect of losing a player. Rows (3) and (8) are, effectively,

measuring the team's new equilibrium.

The volatility moderation results for Game Score are not sensitive to the minimum absence length. Moving from a minimum of 5 games to either 7 or 10 produces a coefficient with similar magnitude and significance. When instead using Box Plus Minus, the longer minimum absence produces results that are not statistically significant for Win Rate. In the main specification, this is the least-significant cell, so the attenuation is not unexpected.

Table G.1: Alternative specifications. Two-step standard errors, clustered by team-season, in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

	<i>Game Score</i>			<i>Box Plus Minus</i>		
	τ	β_{μ}	β_{σ}	τ	β_{μ}	β_{σ}
<i>Point differential</i>						
(1) Main specification	-0.974*** (0.337)	-0.169* (0.090)	+0.612*** (0.189)	-0.974*** (0.334)	-0.319** (0.136)	+0.380** (0.154)
(2) Only games missed, within the window	-0.780** (0.366)	-0.240** (0.105)	+0.708*** (0.207)	-0.780** (0.367)	-0.332** (0.154)	+0.389** (0.172)
(3) All games missed (no window)	-0.132 (0.419)	-0.336*** (0.101)	+0.331 (0.218)	-0.132 (0.403)	-0.573*** (0.129)	-0.008 (0.170)
(4) Absence is 7+ games	-0.934*** (0.337)	-0.177* (0.104)	+0.519** (0.208)	-0.934*** (0.334)	-0.348** (0.155)	+0.334** (0.160)
(5) Absence is 10+ games	-0.651* (0.339)	-0.182* (0.093)	+0.668*** (0.197)	-0.651* (0.340)	-0.339** (0.152)	+0.321** (0.159)
<i>Win rate</i>						
(6) Main specification	-0.022* (0.012)	-0.0052* (0.0031)	+0.0222*** (0.0072)	-0.022* (0.012)	-0.0088* (0.0046)	+0.0124** (0.0055)
(7) Only games missed, within the window	-0.021 (0.013)	-0.0053 (0.0036)	+0.0225*** (0.0081)	-0.021 (0.013)	-0.0070 (0.0051)	+0.0105* (0.0062)
(8) All games missed (no window)	-0.003 (0.015)	-0.0083** (0.0033)	+0.0140 (0.0086)	-0.003 (0.015)	-0.0132*** (0.0045)	-0.0014 (0.0063)
(9) Absence is 7+ games	-0.029** (0.012)	-0.0048 (0.0035)	+0.0158** (0.0076)	-0.029** (0.012)	-0.0092* (0.0051)	+0.0069 (0.0055)
(10) Absence is 10+ games	-0.017 (0.012)	-0.0054* (0.0031)	+0.0194*** (0.0072)	-0.017 (0.012)	-0.0086* (0.0049)	+0.0070 (0.0054)

APPENDIX H: ENDOGENOUS ABSENCES

A material fraction of qualifying absences (37 of 268) are not injuries, but cases where a player is removed from the roster and, usually, traded to another team. Unlike injuries, which are quasi-random, trades are strategic moves. I have left these events in the main specification for two reasons. First, the timing of a trade requires both the focal organization and one or more counter-parties to simultaneously agree to the transaction and the value of the included player(s). Further, the NBA's salary-matching rules further ensure that the wages of the traded players are similar. An organization cannot unilaterally make a trade and so the timing is still, to some extent, outside their control. Second, for purposes of team-building any strategic behavior would be expected to improve the team's position. This would bias the treatment effect to zero while still leaving useful variation with which to estimate the moderating effect of volatility.

Still, to ensure that endogenous roster choices are not driving observed results, I re-estimate the manuscript's results in Table 5 with these 37 events omitted. As expected, the observed effects are stronger. Trades do, in expectation, improve the transacting organization. When I remove trades, the average cost of an absence increases substantially and the effect of volatility is substantially unchanged.

Table H.1: Departures. Rows (2) and (4) drop the 37 events in which the absent player appears for another franchise later the same season, leaving 231 events. Rows (1) and (3) reproduce the Table 5 results. Two-step standard errors, clustered by team-season, in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

	<i>Game Score</i>			<i>Box Plus Minus</i>		
	τ	β_{μ}	β_{σ}	τ	β_{μ}	β_{σ}
<i>Point differential</i>						
(1) Main specification	-0.974*** (0.337)	-0.169* (0.090)	+0.612*** (0.189)	-0.974*** (0.334)	-0.319** (0.136)	+0.380** (0.154)
(2) Excluding departures	-1.224*** (0.335)	-0.256*** (0.080)	+0.637*** (0.205)	-1.224*** (0.330)	-0.422*** (0.117)	+0.375** (0.165)
<i>Win rate</i>						
(3) Main specification	-0.022* (0.012)	-0.0052* (0.0031)	+0.0222*** (0.0072)	-0.022* (0.012)	-0.0088* (0.0046)	+0.0124** (0.0055)
(4) Excluding departures	-0.031*** (0.012)	-0.0078*** (0.0029)	+0.0225*** (0.0079)	-0.031*** (0.012)	-0.0118*** (0.0041)	+0.0114* (0.0058)